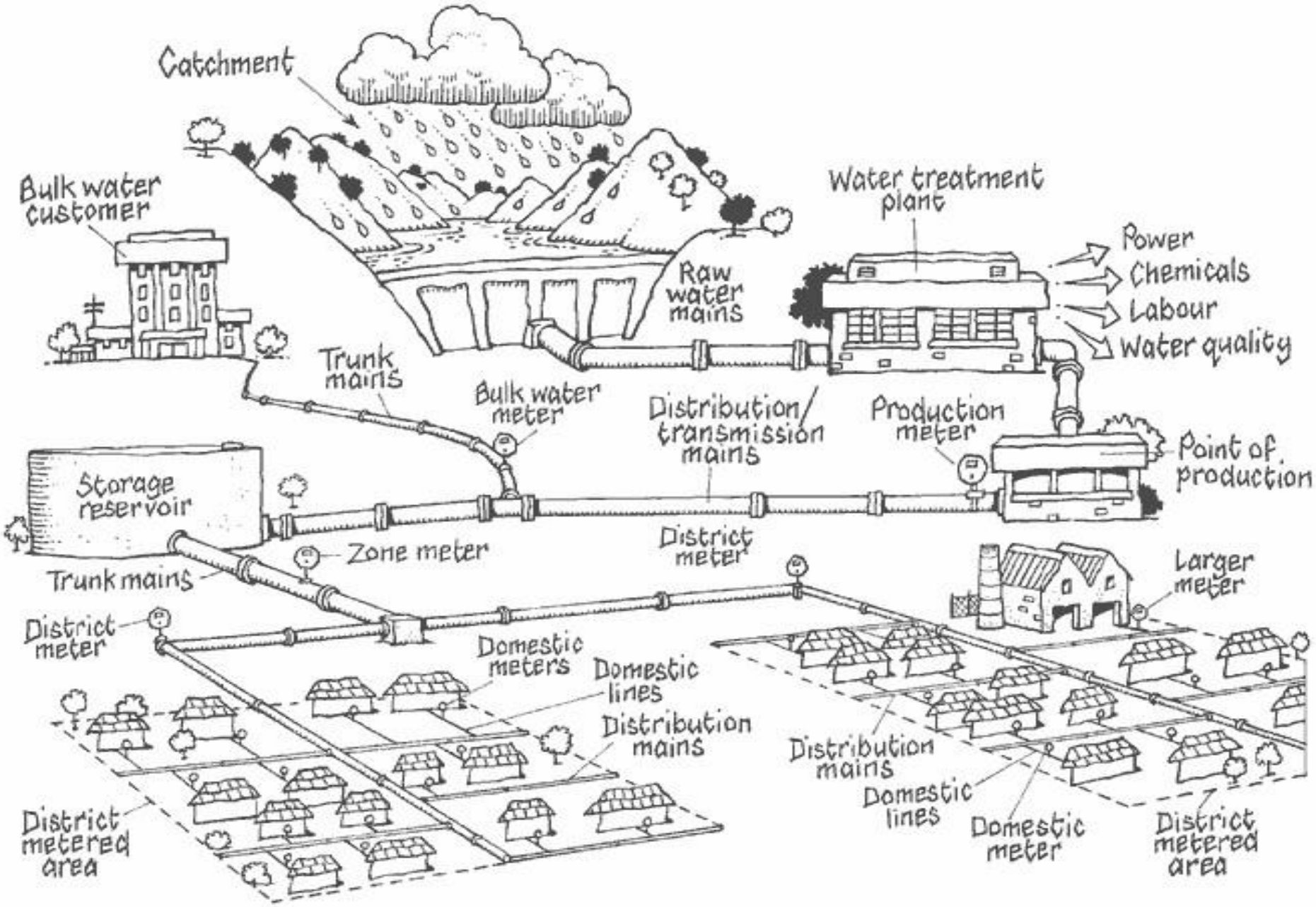
Content

- Water Distribution Systems
- Design of Water Distribution Systems
- Pipe Network Analysis

Why Treat Water?

- *Society realized long ago that human health and the welfare of the general population are improved if public water supplies are treated prior to use.*
- Nearly all structures require a water supply.
- Appropriate flow rate, pressure, and water quality are necessary for effective use.

WATER DISTRIBUTION SYSTEM



Part B

Design of Water Distribution Systems

Water Distribution Systems

Introduction

To deliver water to individual consumers with appropriate quality, quantity, and pressure in a community setting requires an extensive system of:

Pipes.

Storage reservoirs.

Pumps.

Other related accessories.

Distribution system: is used to describe collectively the facilities used to supply water from its **source** to the **point of usage** .

Methods of Supplying Water

- Depending on the topography relationship between the source of supply and the consumer, water can be transported by:
  - Canals.
  - Tunnels.
  - Pipelines.
- The most common methods are:
  - Gravity supply
  - Pumped supply
  - Combined supply

Design of Water Distribution Systems

A **properly designed** water distribution system should fulfill the following requirements:

Main requirements :

- Satisfied quality and quantity standards

Additional requirements :

- To enable reliable operation during irregular situations (power failure, fires..)
- To be economically and financially viable, ensuring income for operation, maintenance and extension.
- To be flexible with respect to the future extensions.

Design of Water Distribution Systems

The design of water distribution systems must undergo through different studies and steps:

Design Phases

Preliminary Studies

Network Layout

Hydraulic Analysis

Preliminary Studies:

Must be performed before starting the actual design:

4.3.A.1 Topographical Studies:

1. Contour lines (or controlling elevations).
2. Digital maps showing present (and future) houses, streets, lots, and so on..
3. Location of water sources so to help locating distribution reservoirs.

Water Demand Studies:

Water consumption is ordinarily divided into the following categories:

- ☑ Domestic demand.
- ☑ Industrial and Commercial demand.
- ☑ Agricultural demand.
- ☑ Fire demand.
- ☑ Leakage and Losses.



Domestic demand

- It is the amount of water used for Drinking, Cocking, Gardening, Car Washing, Bathing, Laundry, Dish Washing, and Toilet Flushing.
- The average water consumption is different from one population to another. In Gaza strip the average consumption is 70 L/capita/day which is very low compared with other countries. For example, it is 250 L/c/day in United States, and it is 180 L/c/day for population live in Cairo (Egypt).
- The average consumption may increase with the increase in standard of living.
- The water consumption varies hourly, daily, and monthly

Design of Water Distribution Systems

The total amount of water for domestic use is a function of:

Population increase

How to predict the increase of population?

Use  Geometric-increase model

$$P = P_0 (1+r)^n$$

P_0 = recent population
r = rate of population growth
n = design period in years
P = population at the end of the design period.

The total domestic demand can be estimated using:

$$Q_{domestic} = Q_{avg} * P$$

Industrial and Commercial demand

- It is the amount of water needed for factories, offices, and stores….
- Varies from one city to another and from one country to another
- Hence should be studied for each case separately.
- However, it is sometimes taken as a percentage of the domestic demand.

Agricultural demand

- It depends on the type of crops, soil, climate…

Fire demand

- To resist fire, the network should save a certain amount of water.
- Many formulas can be used to estimate the amount of water needed for fire.

Design of Water Distribution Systems

Fire demand Formulas

$$Q_F = 65\sqrt{P}(1 - 0.01\sqrt{P})$$

Q_F = fire demand l/s
P = population in thousands

$$Q_F = 53\sqrt{P}$$

Q_F = fire demand l/s
P = population in thousands

$$Q_F = 320 * C\sqrt{A}$$

Q_F = fire demand flow m^3/d
A = areas of all stories of the building under consideration (m^2)
C = constant depending on the type of construction;

The above formulas can be replaced with local ones
(Amounts of water needed for fire in these formulas are high).

Leakage and Losses

- This is " unaccounted for water "(UFW)
- It is attributable to:

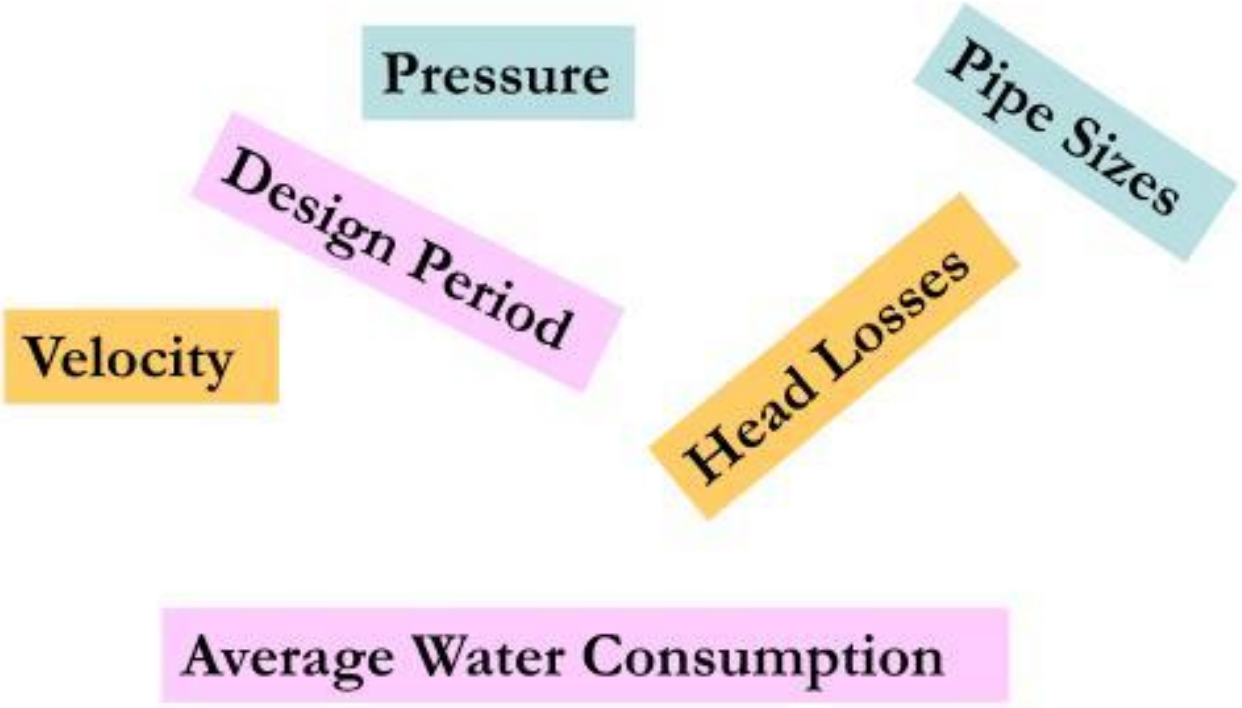
Errors in meter readings

Unauthorized connections

Leaks in the distribution system

Design Criteria

Are the design limitations required to get the most efficient and economical water-distribution network



Velocity

- Not be lower than 0.6 m/s to prevent sedimentation
- Not be more than 3 m/s to prevent erosion and high head losses.
- Commonly used values are 1 - 1.5 m/sec.

Design of Water Distribution Systems

Pressure

- Pressure in municipal distribution systems ranges from 150-300 kPa in residential districts with structures of four stories or less and 400-500 kPa in commercial districts.

- Also, for fire hydrants the pressure should not be less than 150 kPa (15 m of water).

- In general for any node in the network the pressure should not be less than 25 m of water.

- Moreover, the maximum pressure should be limited to 70 m of water

Design of Water Distribution Systems

Pipe sizes

- Lines which provide only domestic flow may be as small as 100 mm (4 in) but should not exceed 400 m in length (if dead-ended) or 600 m if connected to the system at both ends.

- Lines as small as 50-75 mm (2-3 in) are sometimes used in small communities with length not to exceed 100 m (if dead-ended) or 200 m if connected at both ends.

- The size of the small distribution mains is seldom less than 150 mm (6 in) with cross mains located at intervals not more than 180 m.

- In high-value districts the minimum size is 200 mm (8 in) with cross-mains at the same maximum spacing. Major streets are provided with lines not less than 305 mm (12 in) in diameter.

Head Losses

- Optimum range is 1-4 m/km.
- Maximum head loss should not exceed 10 m/km.

Design Period for Water supply Components

- The economic design period of the components of a distribution system depends on
  - Their life.
  - First cost.
  - And the ease of expandability.

Average Water Consumption

- From the water demand (preliminary) studies, estimate the average and peak water consumption for the area.

INTAKE STRUCTURES

WHAT IS AN INTAKE STRUCTURE ?

- ***The basic function of intake structure is to help in safely withdrawing water from the source and then to discharge this water in to the withdrawal conduit, through which it reaches the water treatment plant.***
- It is constructed at the entrance of the withdrawal conduit and thereby protecting it from being damaged/clogged by ice, debris.
- Some times from reservoirs where gravity flow is possible, water is directly transmitted to the treatment through intake structure.
- If gravity flow is not possible, water entering intake structure is lifted by pumps and taken to the treatment plant.

SELECTING LOCATION OF INTAKE STRUCTURE

- Site should be near the treatment plant to reduce conveyance cost.
- Intake must be located in the purer zone of the source so that best quality water is withdrawn from source to reduce the load on the treatment plant.
- Intake must never be located in the vicinity of waste water disposal point.
- Intake must never be located near the navigation channels so as to reduce chances of pollution due to waste discharge from ships.
- The site should be such as to permit greater withdrawal of water, if required in future.

SELECTING LOCATION OF INTAKE STRUCTURE

- Intake must be located at a place from where it can draw water even during the driest period of the year.
- The intake site should remain easily accessible during floods and should not get flooded.
- In meandering rivers, the intakes should not be located on curves or atleast on sharp curves.

Water Intake Structures

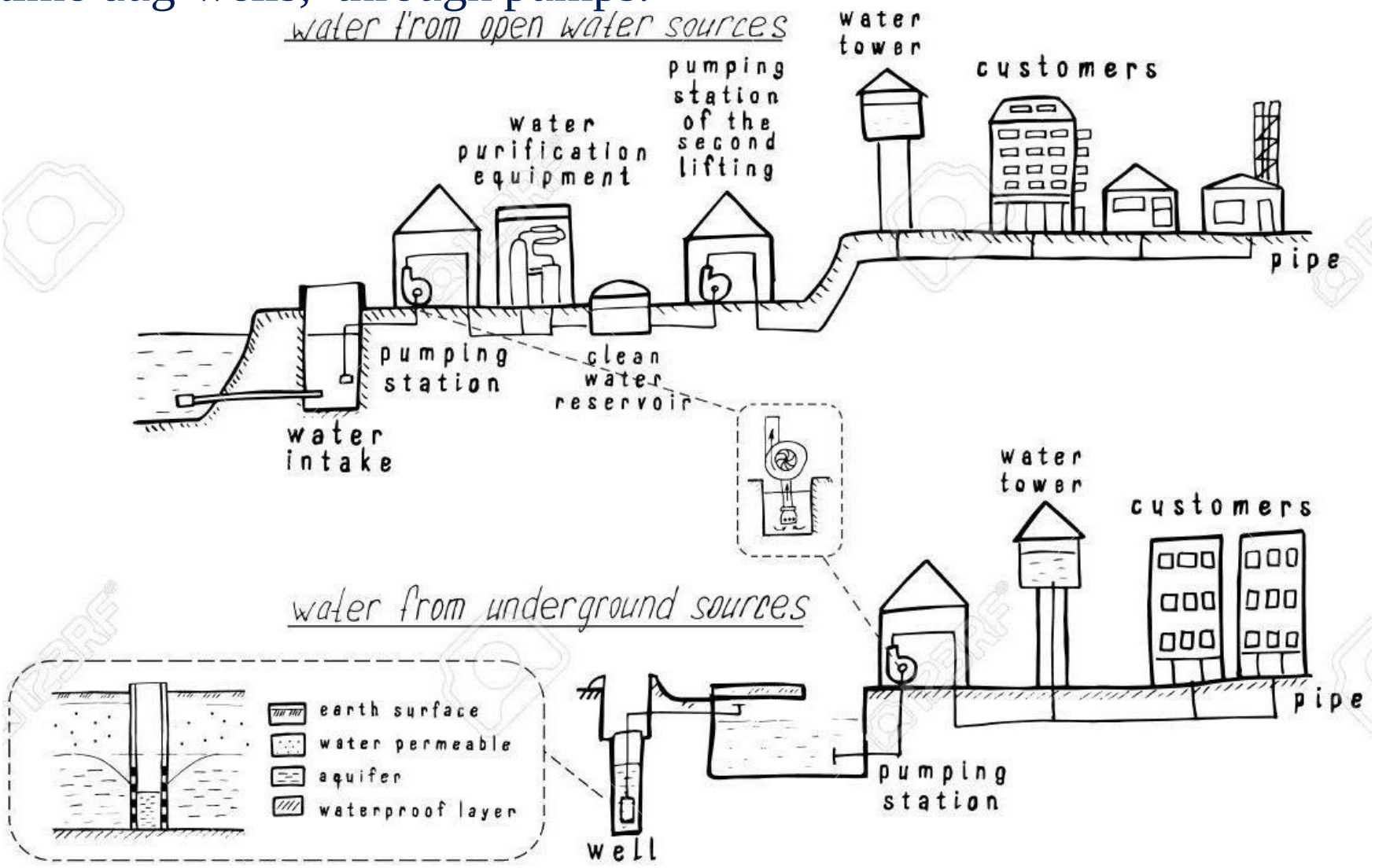
- Water Intake structures are systems installed to help in safely withdrawing water from the source and then to discharge this water in to the withdrawal (or intake) conduit, through which it reaches the water treatment plant.
- Two approaches:
  - ***Pumping***
  - ***Conveyance***



Image Source: https://www.gkw-consult.com/en/projects/water/single/article/india-kerala-water-supply-project/

Types of Intake

There are various types of intake structures for surface water sources, while groundwater are commonly withdrawn through bore- wells, or at time dug-wells, through pumps.



TYPES OF INTAKES

- Types of Intakes

According to type of source

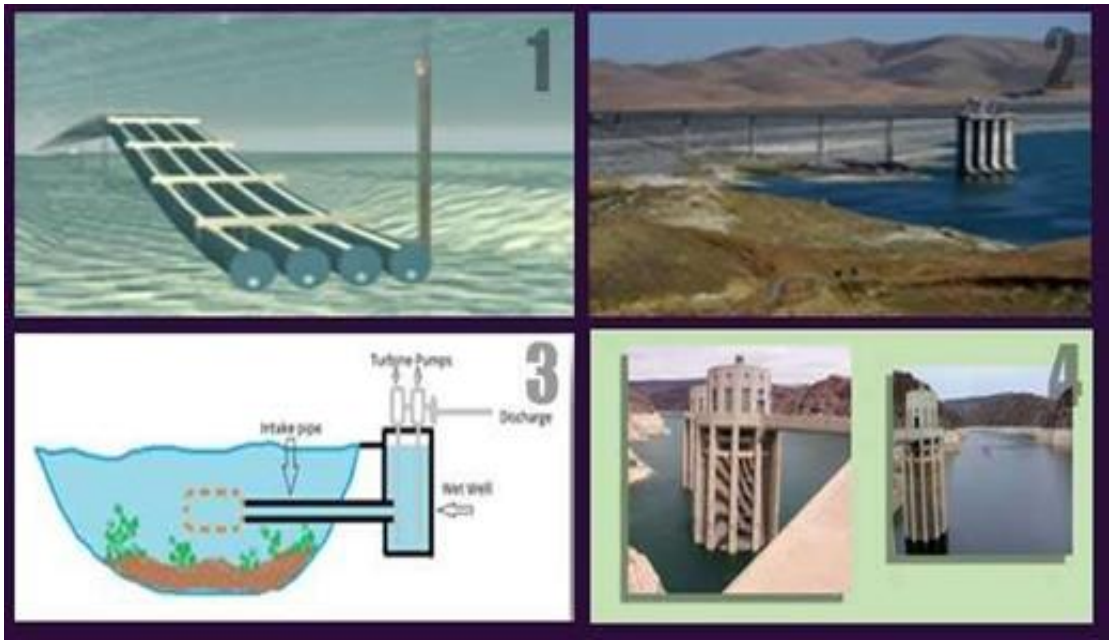
- River Intake
- Canal Intake
- Reservoir Intake
- Lake Intake

According to position of Intake

- Submerged Intake
- Exposed Intake

According to presence of water in the tower

- Wet Intake
- Dry Intake



- Lake Intake are mostly submerged intake. These Intakes are constructed in the bed of lake below the low water level so as to draw water even in dry season. It mainly consist of a pipe laid in the bed of the lake.
- One end of the pipe which is in middle of the lake is fitted with bell mouth opening covered with a mesh and protected timber or concrete crib.
- The water enters in the pipe through the bell mouth opening and flows under gravity to the bank where it is collected in a sump well and then pumped to the treatment plant for necessary treatment.
- According to position of Intake

(a) Submerged Intake

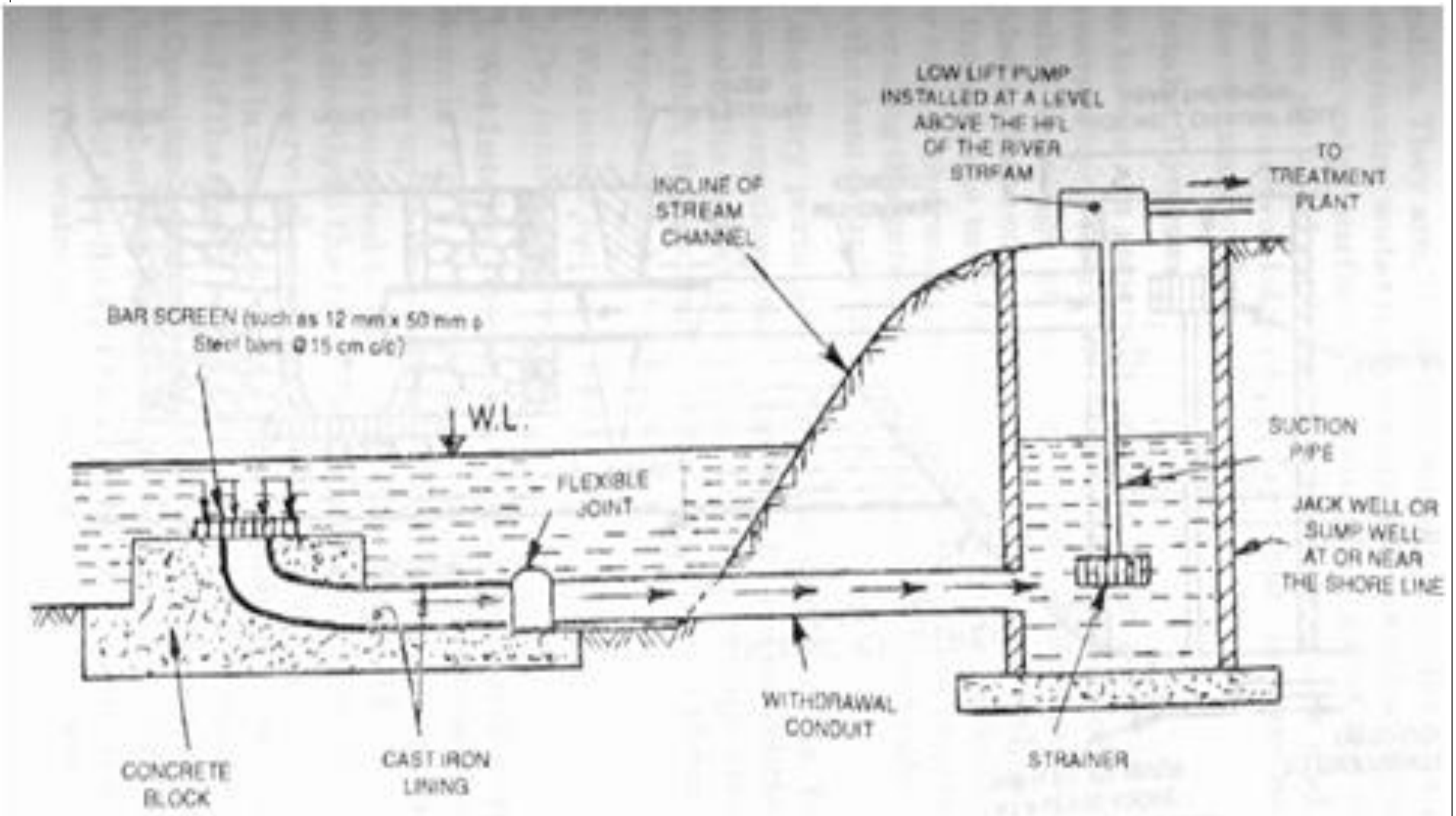
(b) Exposed Intake

- The submerged Intake structures are those which are constructed entirely under water. They are less expensive to construct but are difficult to maintain.
- Such intakes are commonly used to obtain water from lakes The Exposed intakes is in the form of well or tower constructed near the bank of river or in some cases even away from the bank of river.
- They are more common due to ease in operation and maintenance

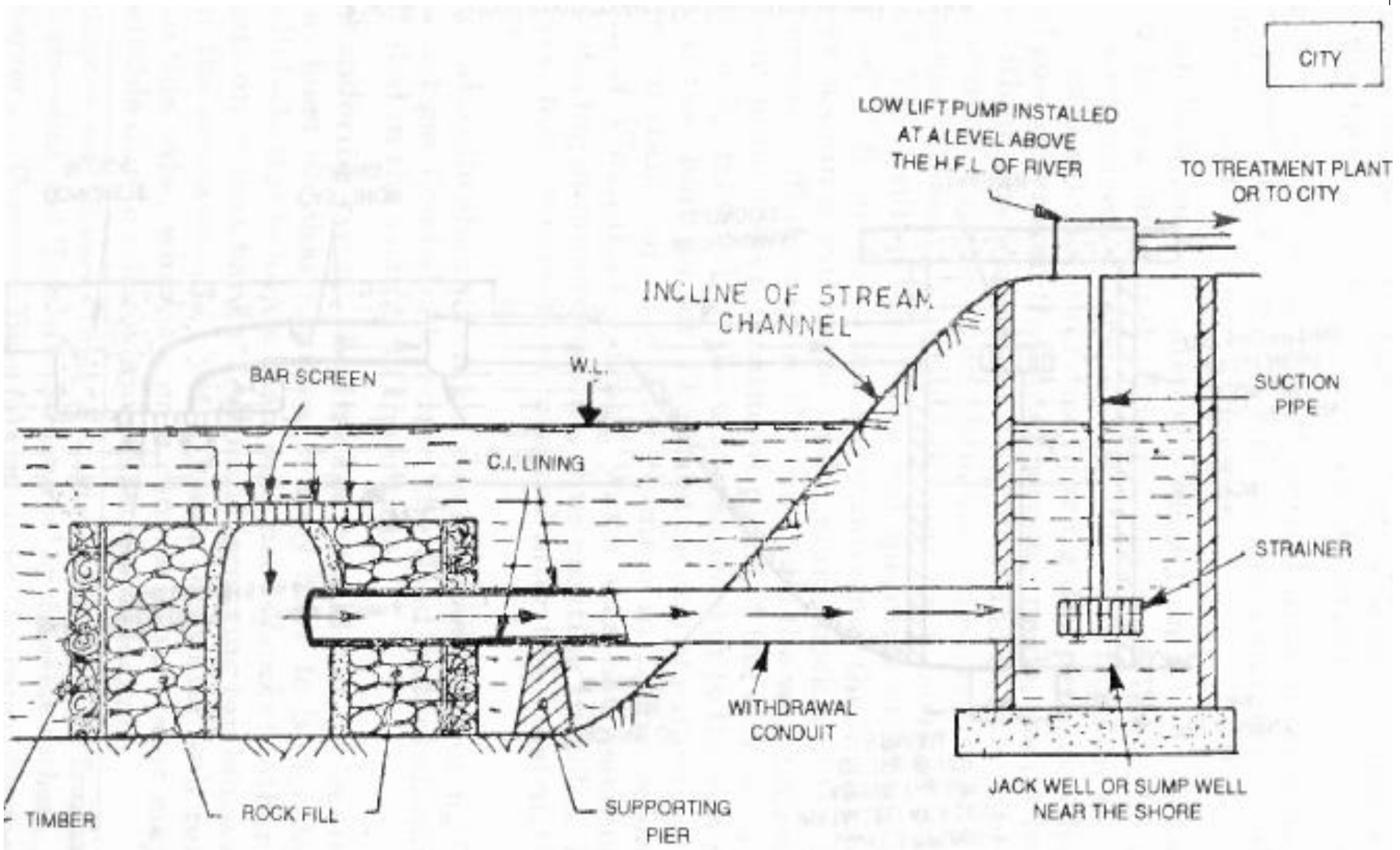
SIMPLE SUBMERGED INTAKES

- It consists of a simple concrete block or a rock filled timber crib supporting the starting end of the withdrawal pipe.
- The intake opening is generally covered by screen so as to prevent the entry of debris, ice etc. in to the withdrawal conduit.
- In lakes, where silt tends to settle down , the intake opening is generally kept at about 2 to 2.5m above the lake bed level to avoid entry of silt.
- They are cheap & do not obstruct navigation
- They are widely used for small water supply projects drawing water from streams or lakes having a little change in water level through out year.
- Limitation is that they are not easily accessible for cleaning & repairing.

SIMPLE SUBMERGED INTAKES



ROCK FILLED TIMBER CRIB - SUBMERGED INTAKE



Types of Intakes According to presence of Water in the tower

- They are widely used on large water supply projects drawing water from rivers or reservoirs having large change in water level.

- Gate controlled openings called ***Ports*** are provided at various levels in these concrete towers to regulate the flow.

- If the entry ports are submerged at all levels, there is no problem of any clogging or damage by ice or debris etc.

- There are two major types of intake towers:

  (a) Wet intake towers

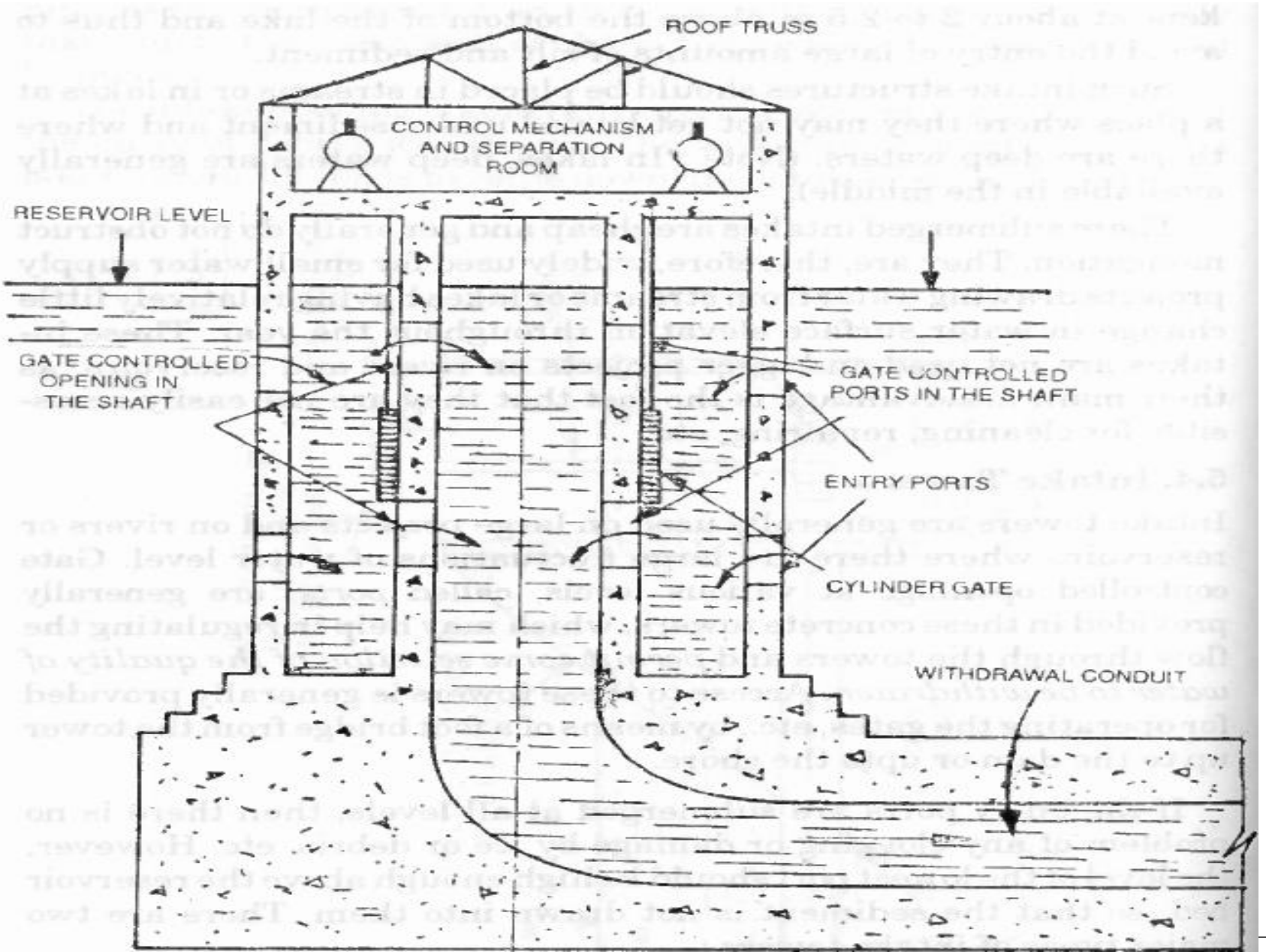
  (b) Dry intake towers

  (c) Movable intake towers

WET INTAKE TOWERS

- It consist of a concrete circular shell filled with water up to the reservoir level and has a vertical inside shaft which is connected to the withdrawal pipe.
- The withdrawal pipe may lie over the bed of the rivers or may be in the form of tunnels below the river bed.
- Openings are made in to the outer concrete shell as well as, in to the inside shaft.
- Gates are usually placed on the shaft, so as to control the flow of water in to the shaft and the withdrawal conduit.
- The water coming out of the withdrawal pipe may be taken to pump house for lift (if treatment plant is at high elevation) or may be directly taken to treatment plant (at lower elevation).

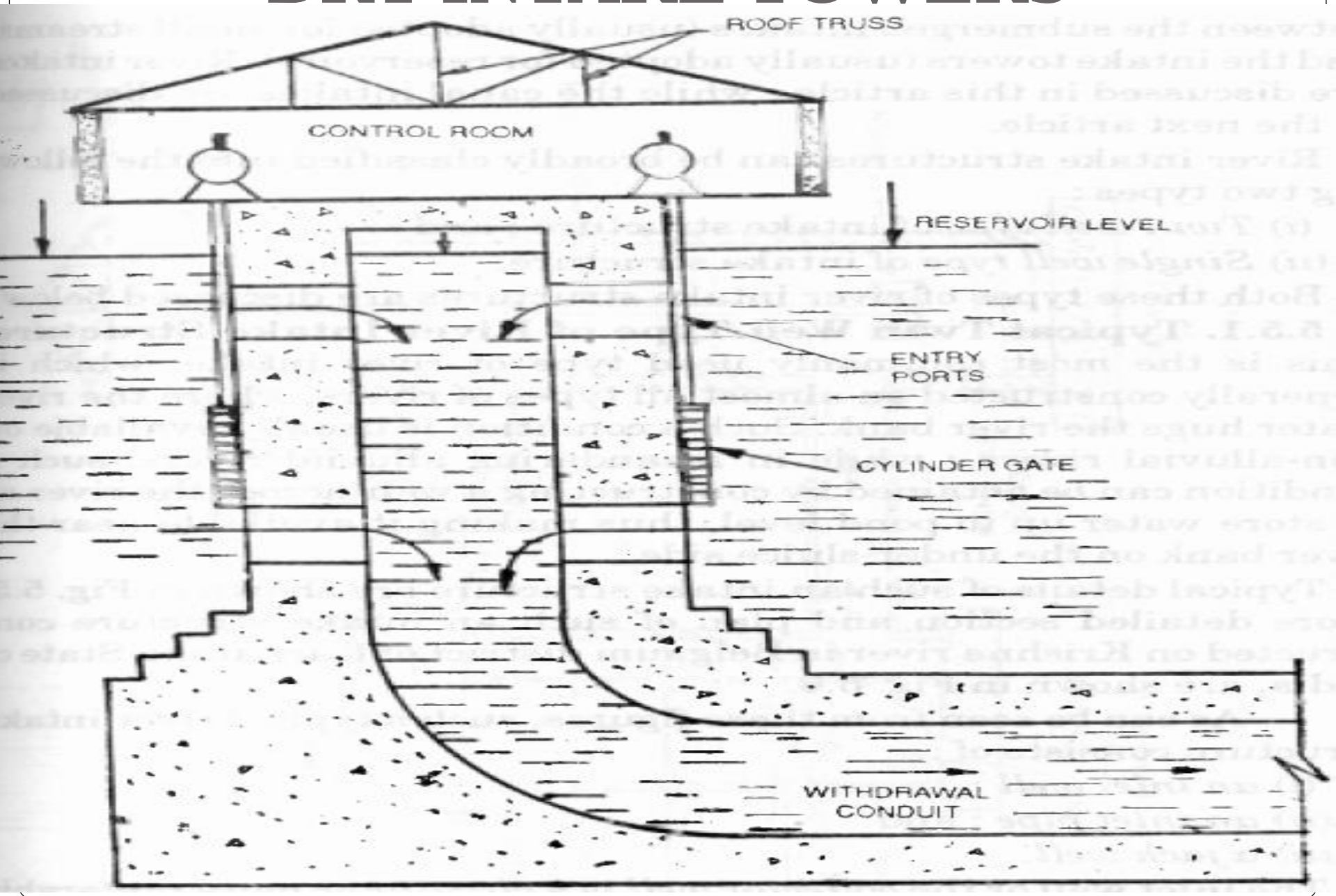
WET INTAKE TOWERS



DRY INTAKE TOWERS

- The water is directly drawn in to the withdrawal conduit through the gated entry ports.
- It has no water inside the tower if its gates are closed.
- When the entry ports are closed, a dry intake tower will be subjected to additional buoyant forces.
- Hence it must be of heavier construction than wet intake tower.
- They are useful since water can be withdrawn from any selected level of the reservoir by opening the port at that level.

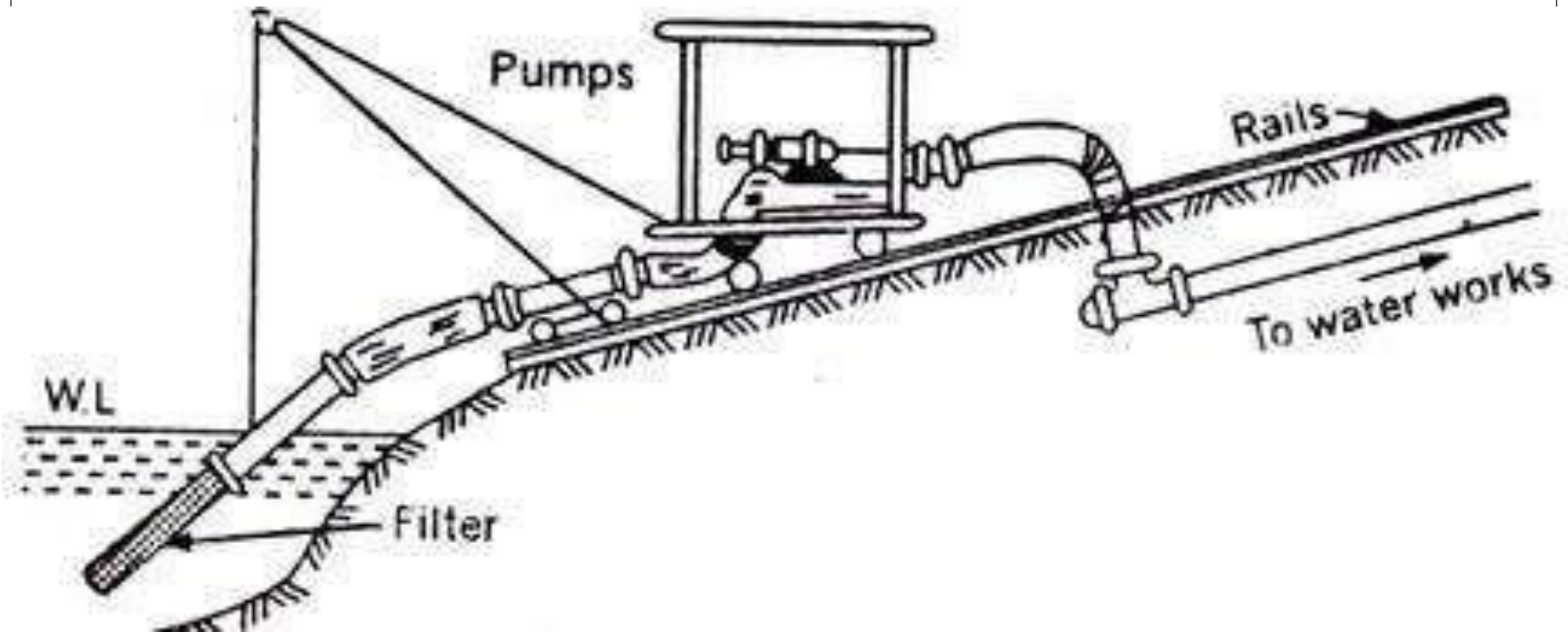
DRY INTAKE TOWERS





Movable Intake

- In case of emergency and temporary works, movable intakes can be used.
- Pumping plant is installed in a carriage or trolley and the suction pipe having strainer pipe at the end is lowered in the water.
- The water is directly pumped from the river and sent for the treatment / distribution.



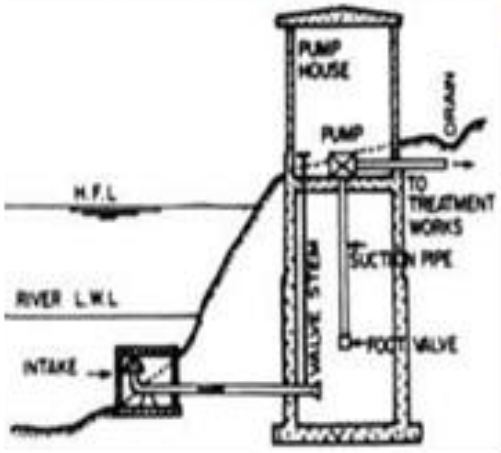
Types of Intakes
According to type of source

River Intake

- A River Intake is located on the upstream side of the city to get comparatively better quality of water.
- They are either located sufficiently inside the river so that necessary demand of water can be met in all the seasons of the year.
- The intake tower permits the entry of water through several entry ports located at various levels to cope with fluctuations in the water levels during different seasons. This are also called as penstocks.
- The penstocks are covered with suitable design screens to prevent entry of floating impurities.

River Intake

- It is a type of intake which may either located sufficiently inside the river so that demands of water are met with in all the seasons of the year, or they may be located near the river bank where a sufficient depth of water is available.
- Sometimes, an approach channel is constructed and water is led to the intake structure.



- If the water level in the river is low, a weir may be constructed across it to raise the water level and divert it to the intake structure.



Image Source: https://theconstructor.org/structures/intake-structures-types-of-intakes/11233/

Design Consideration for River Intake Structures

Criteria	Important for Water Supply Criteria	Important for Aquatic Life
Structural Stability	Structure should be structurally sound and not damaged by floods	Structure should not promote bank erosion or other forms of channel instability
Sustaining Sediment Transport	Should sustain a pool of adequate depth in front of the intake, and should not accumulate sediment and debris	Should sustain the transport of coarse materials along the stream, and any environmentally- damaging activity should be avoided
Migration Path	Not important	Provide a migration path so that both larval and adult stages are able to migrate, both in the upstream and the downstream direction
Minimum Flow	Structure should sustain withdrawals under all conditions	Guarantee continuous release of water downstream to sustain minimum in-stream flows

MEDIUM SIZED RIVER INTAKE STRUCTURES

- They are generally constructed for withdrawing water from almost all rivers.

- They can be classified in to two types

  (1) Twin well type of intake structure

  (2) Single well type of intake structure

TWIN WELL TYPE INTAKES

- They are constructed on almost all types of rivers, where the river water hugs the river bank.
- A typical river intake structure consists of 3 components:

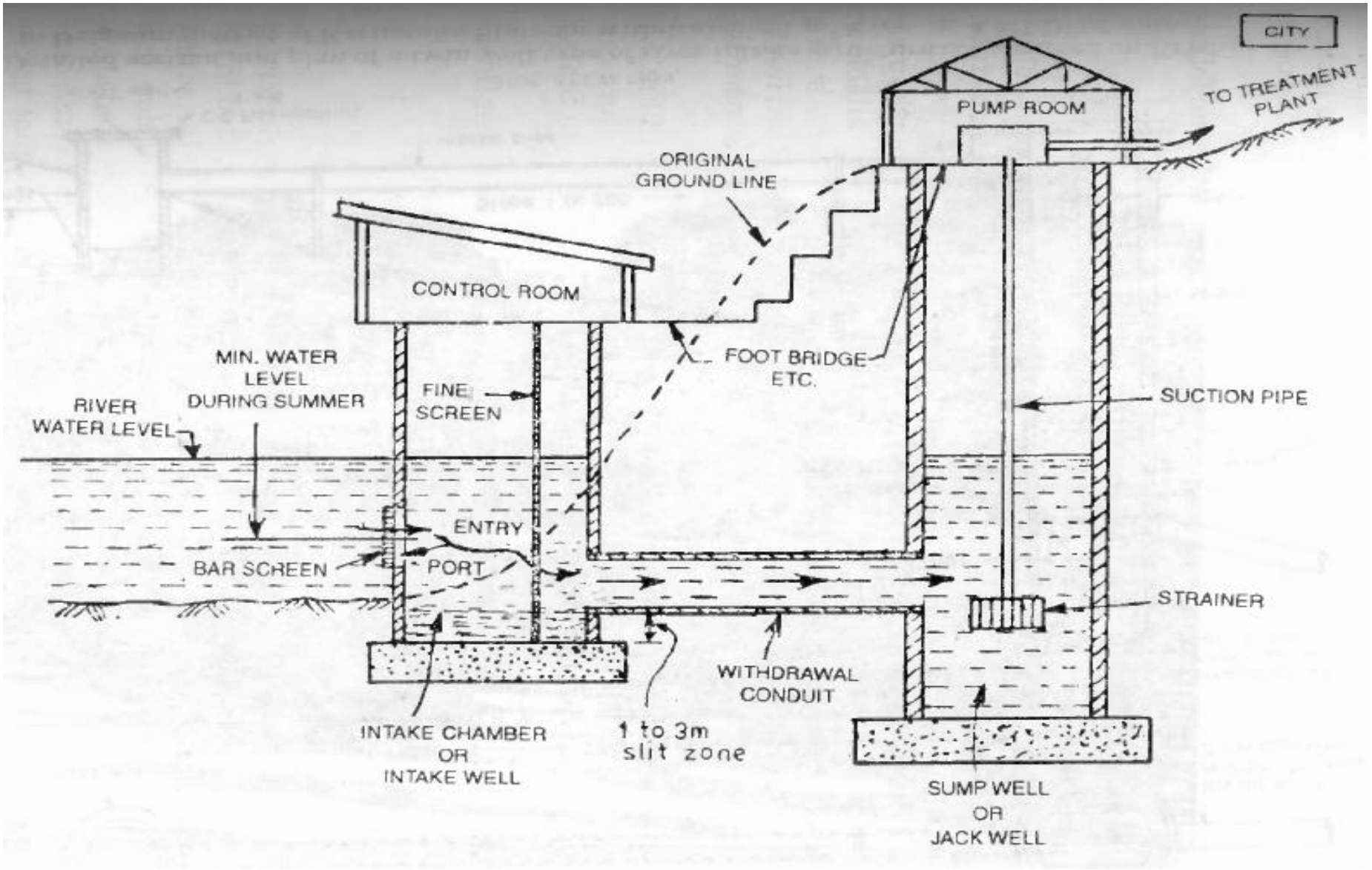
  (a) An inlet well

  (b) An inlet pipe (intake pipe)

  (c) A jack well

- Inlet well is usually circular in c/s, made of masonry or concrete.
- Inlet pipe connects inlet well with jack well. It has a min dia of 45cm, laid at slope of 1 in 200. Flow velocity through it<1.2m/s
- Water entering jack well is lifted by pumps & fed into the rising main
- Jack well should be founded on hard strata having B.C$\geq$ 450 kN/m^2.

TWIN WELL TYPE INTAKES



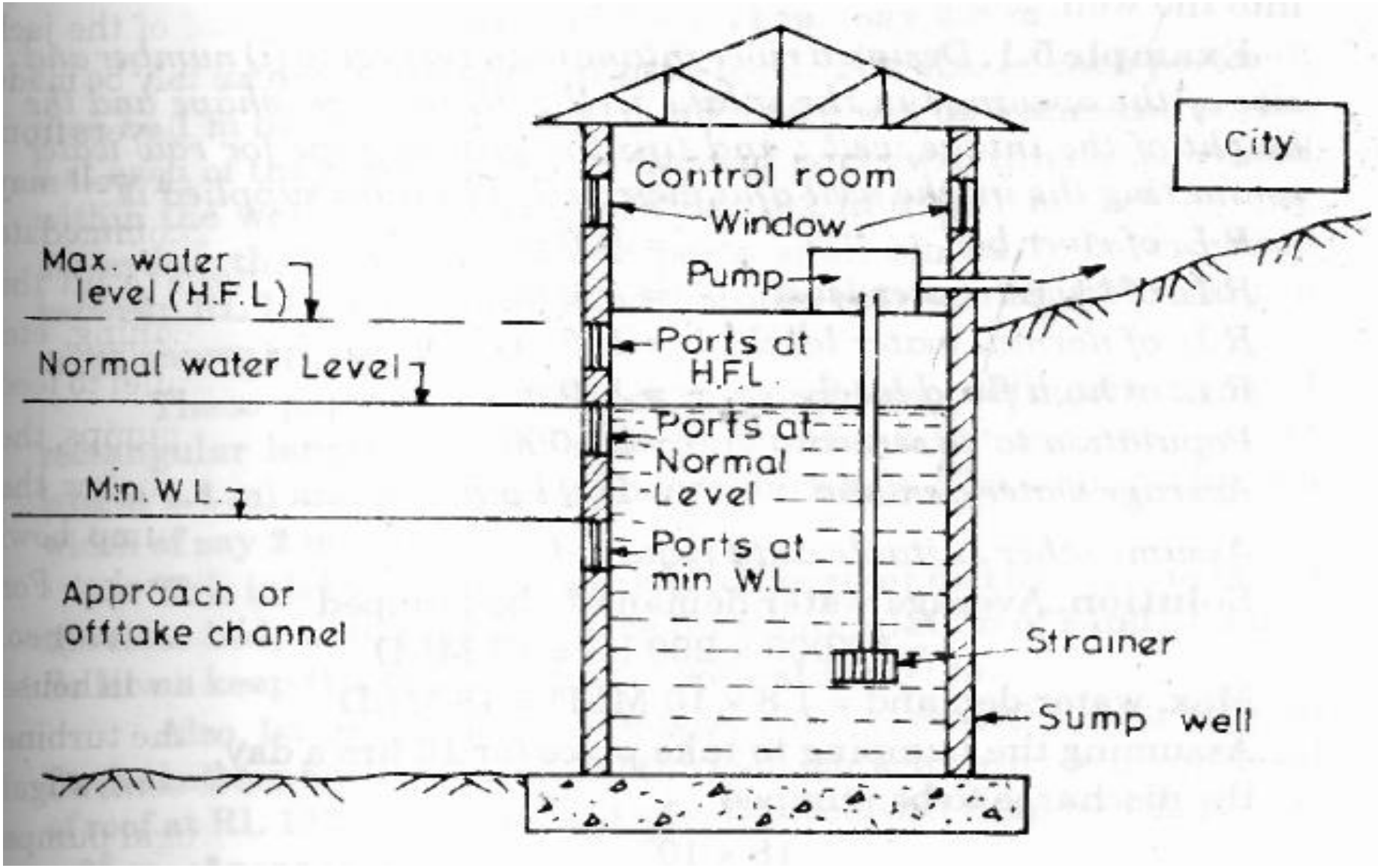
SINGLE WELL TYPE INTAKES

- No inlet well & inlet pipe in this type of river intake.
- Opening or ports fitted with bar screens are provided in the jack well itself.
- The sediment entering will usually be less, since clearer water will enter the off-take channel.
- The silt entering the jack well will partly settle down in the bottom

silt zone of jack well or may be lifted up with the pumped water since pumps can easily lift sedimented water.

- The jack well can be periodically cleaned manually, by stopping the water entry in to the well.

SINGLE WELL TYPE INTAKES





Run-of-River Intakes

- In a run-of-River plants, intake is apparent to power house and draws water from the river without any appreciable storage upstream of the diversion structure.
- Characteristics of river flows., the intake is designed to withstand high peaks and short duration flood flows and high sediment loads.
- The bell mouth entrance is essentially provided with trash racks.



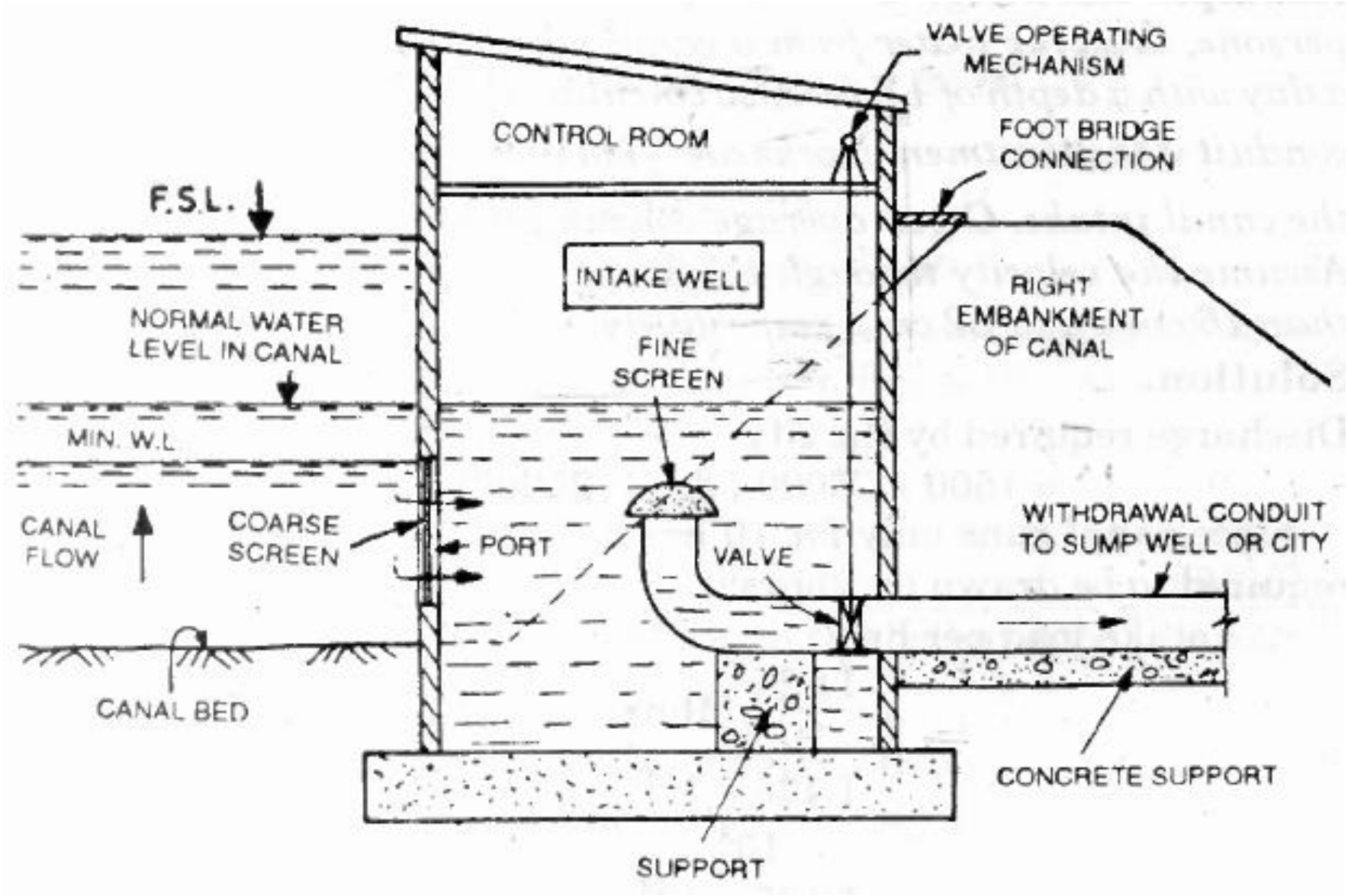
CANAL INTAKES

- In case of a small town a nearby Irrigation Canal can be used as the source of water. The Intake Well is generally located in the bank of the Canal. Since water level is more or less constant there is no need of providing inlets at different depth.
- It essentially consist of concrete or masonry intake chamber or well.
- Since the flow area in the canal is obstructed by the construction of Intake well, the flow velocity in the canal decreases.
- So the canal should be lined on the Upstream & Downstream side of the intake to prevent erosion of sides and bed of channel

- It is also a variant of the run-of-river intake, that is provided adjacent to the diversion weir/ barrages to admit water into the canal.
- It is designed to function under low heads and the topography and geology permits straight reach suitable for it.
- Sediment excluder is an essential component of the intake.
- The crest of the intake is generally raised to prevent entry of coarse fraction of bed load into the canal.



CANAL INTAKES

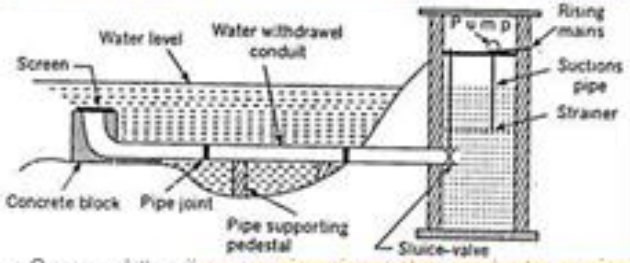


Lake Intake

- Generally submerged intakes, having a pipe laid in the bed of the lake, are preferred for lake intakes.
- One end of the pipe which is in middle of the lake is fitted with bell mouth opening covered with a mesh and protected timber or concrete crib, with crib top covered with cast iron or mesh grating.
- The water enters in the pipe through the bell mouth opening and flows under gravity to the bank where it is collected in a sump well.
- For large supplies, exposed intake towers may also



Image Source: https://theconstructor.org/structures/intake- structures-types-of-intakes/11233/



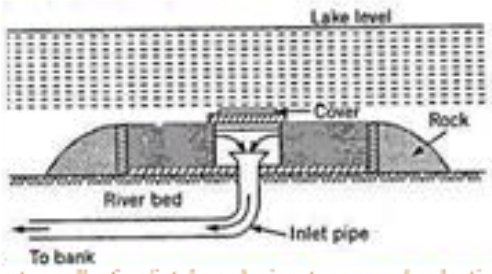
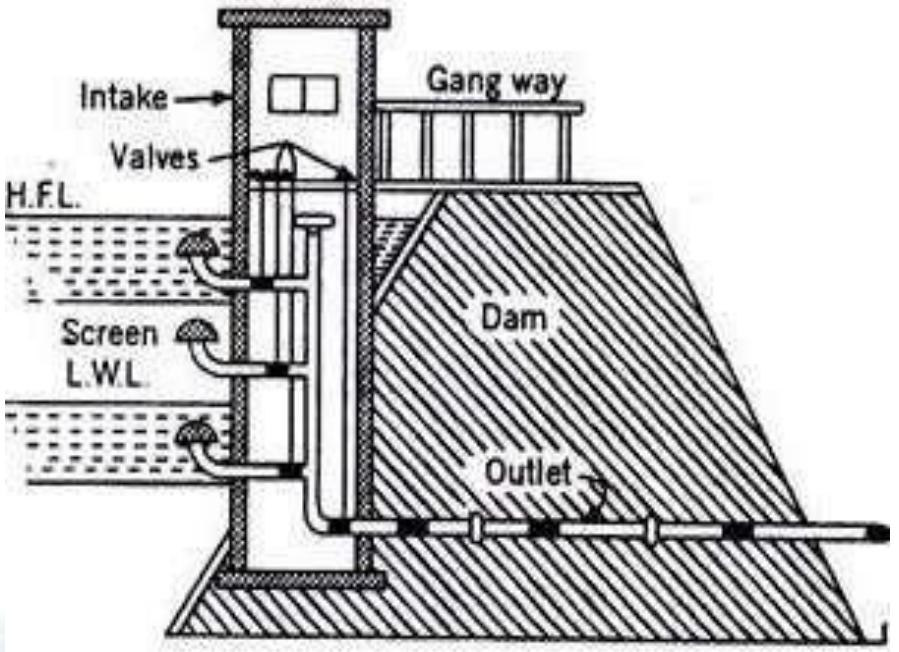


Image Source: https://www.engineeringenotes.com/water-engineering-2/water-collection/intakes-design-types-and-selection-water-collection-water-engineering/16166

Reservoir Intake

- When the flow in the river is not guaranteed throughout the year, a dam is constructed across it to store water in the reservoir so formed. quantity even in the driest period.
- Intake pipes are fixed at different levels, so as to draw water near the surface in all variations of water levels.
- These are similar to river intake, except that these are located near the upstream face of the dam where maximum depth of water is available.
- Design may vary based on the dam type.
- Essentially consists of an intake tower constructed on the slope of dam where Intake can draw water in sufficient





INTAKES FOR SLUICE WAYS OF DAMS

- When the flow in the river is not guaranteed throughout the year, a dam is constructed across the river to store the water in the reservoir so formed.
- Reservoir Intakes essentially consists of an Intake tower constructed on the slope of Dam at such a place where Intake can draw water in sufficient quantity even in the driest period.
- Intake pipes are fixed at different levels, so as to draw water near the surface in all variations of water levels.
- An intake structure constructed at the entrance of conduit and thereby helping in protecting the conduit from being damaged or clogged by ice , trash, debris, etc.., can vary from a simple Concrete block supporting the end of the conduit pipe to huge concrete towers housing intake gates, Screens, pumps, etc.. and even sometimes, living quarters and shops for operating personnel.

INTAKES FOR SLUICE WAYS OF DAMS

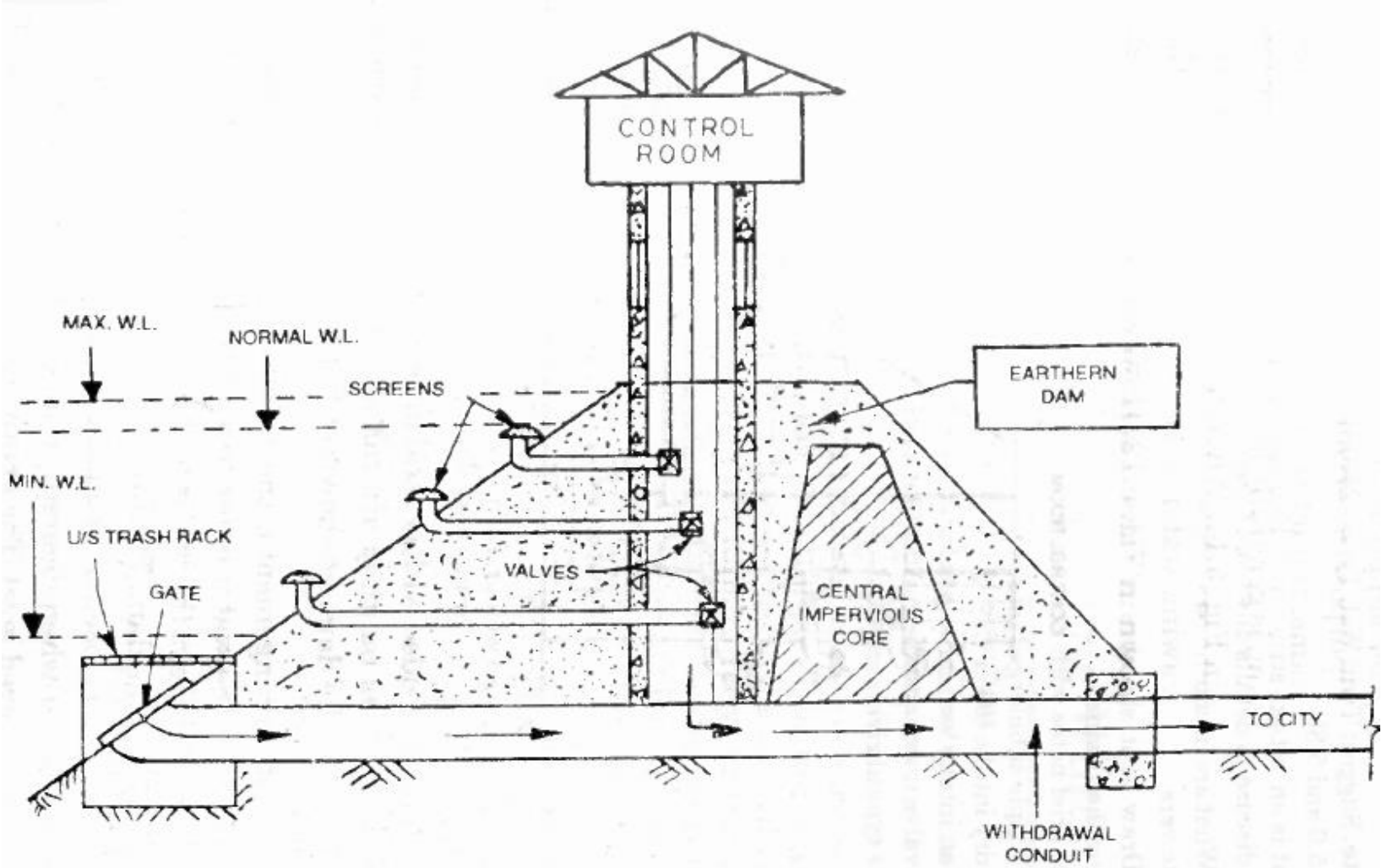


Fig. 5.13. Valve Tower situated within an earthen dam.

INTAKES FOR SLUICE WAYS OF DAMS

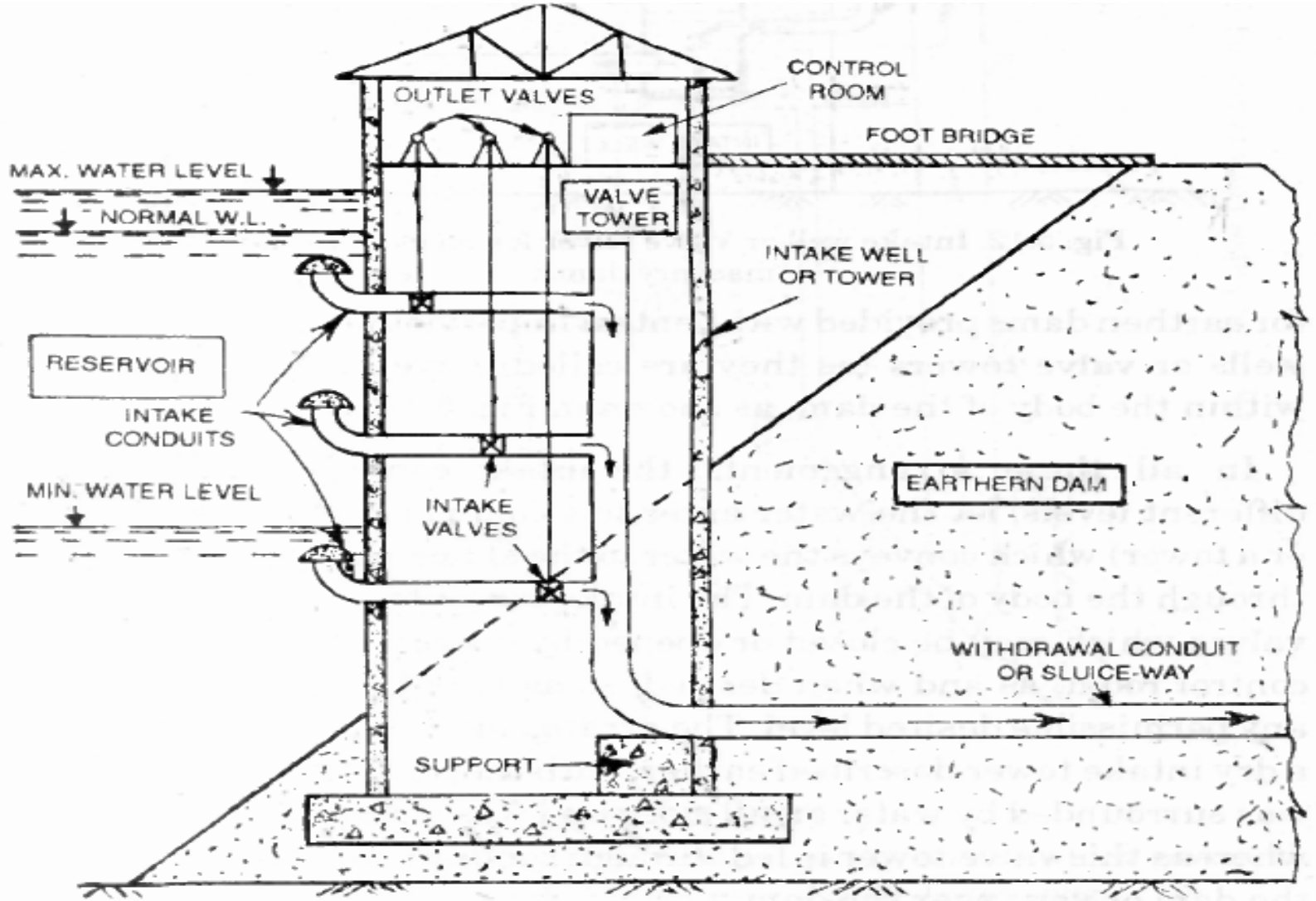


Fig. 5.11. Valve tower situated at the upstream toe of an earthen dam.

INTAKES FOR SLUICE WAYS OF DAMS

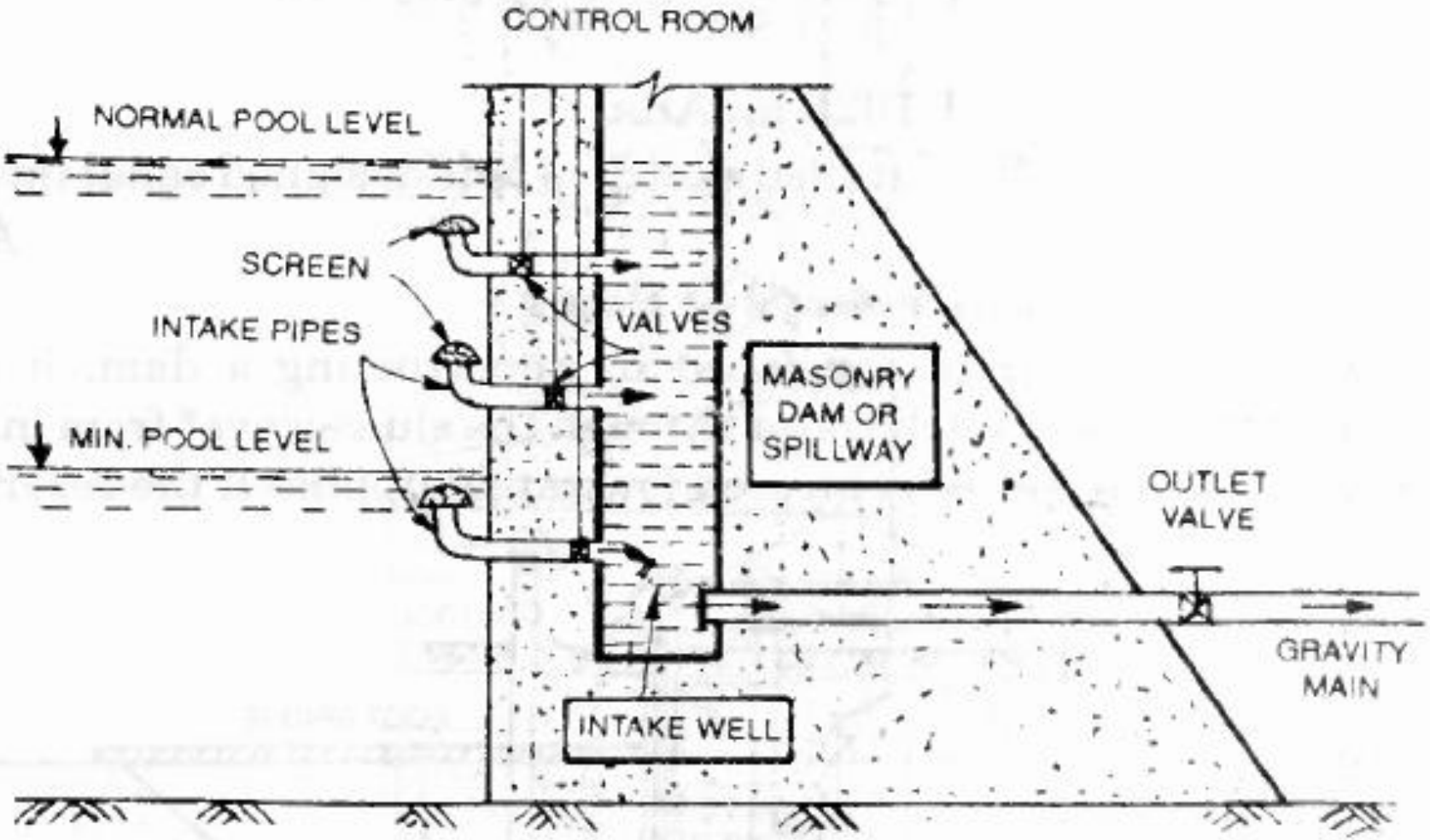


Fig. 5.12. Intake well or Valve tower fcr concrete gravity or masonry dams.

Distribution Reservoirs...

- Distribution reservoirs, also called service reservoirs, are the storage reservoirs, which store the treated water for supplying water during emergencies (such as during fires, repairs, etc.) and also to help in absorbing the hourly fluctuations in the normal water demand.

Functions of Distribution Reservoirs

- to absorb the hourly variations in demand.
- to maintain constant pressure in the distribution mains.
- water stored can be supplied during emergencies.

Location and Height of Distribution Reservoirs

- should be located as close as possible to the centre of demand.
- water level in the reservoir must be at a sufficient elevation to permit gravity flow at an adequate pressure.

Types of Reservoirs...

- Depending upon their elevation w.r.t ground it may be classified into

1. Surface reservoirs
2. Elevated reservoirs
3. Stand Reservoir

SURFACE RESERVOIRS

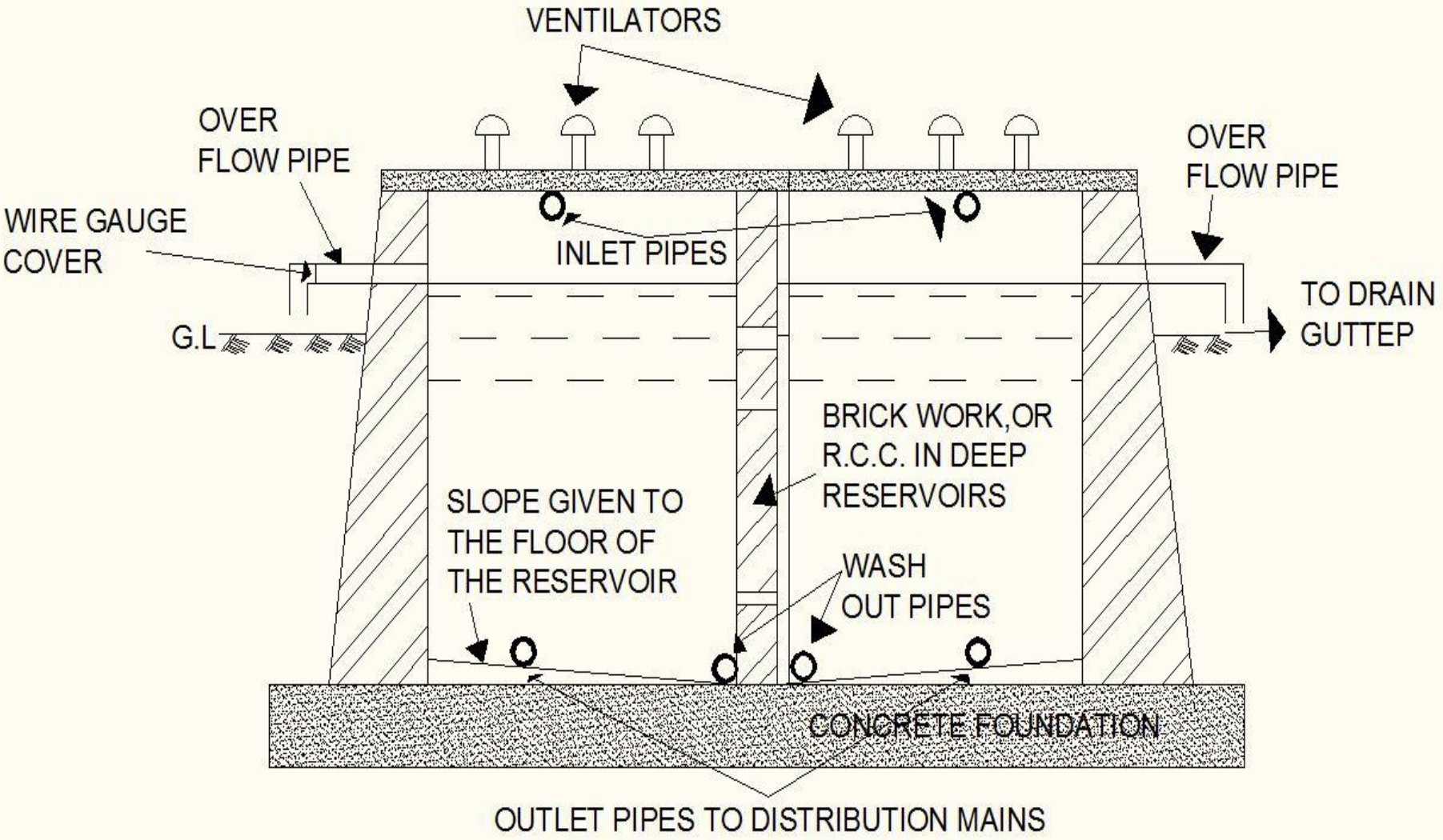
- CONSTRUCTED AT GROUND LEVEL OR BELOW THE GROUND LEVEL.
- GENERALLY CONSTUCTED AT HIGH POINTS IN THE CITY
- MAINLY BASED ON GRAVITATIONAL SYSTEM.
- CHEAP BUT NOT AS GOOD AS ELEVATED RESERVOIRS

ELEVATED RESERVOIRS

- CONSTRUCTED ABOVE THE GROUND LEVEL
- THESE OVER HEAD TANKS ERECTED AT A CERTAIN SUITABLE ELEVATION ABOVE THE GROUND LEVEL AND SUPPORTED ON TOWERS.
- BASED ON COMBINED GRAVITY AND PUMPING SYSTEM.
- CHEAP, EFFICIENT AND RLIABLE

Surface reservoirs…

- These also called ground reservoir.
- Mostly circular or rectangular tank.
- Under ground reservoirs are preferred especially when the size is large.
- These reservoirs are constructed on high natural grounds and are usually made of stones, bricks, plain or reinforced cement concrete.



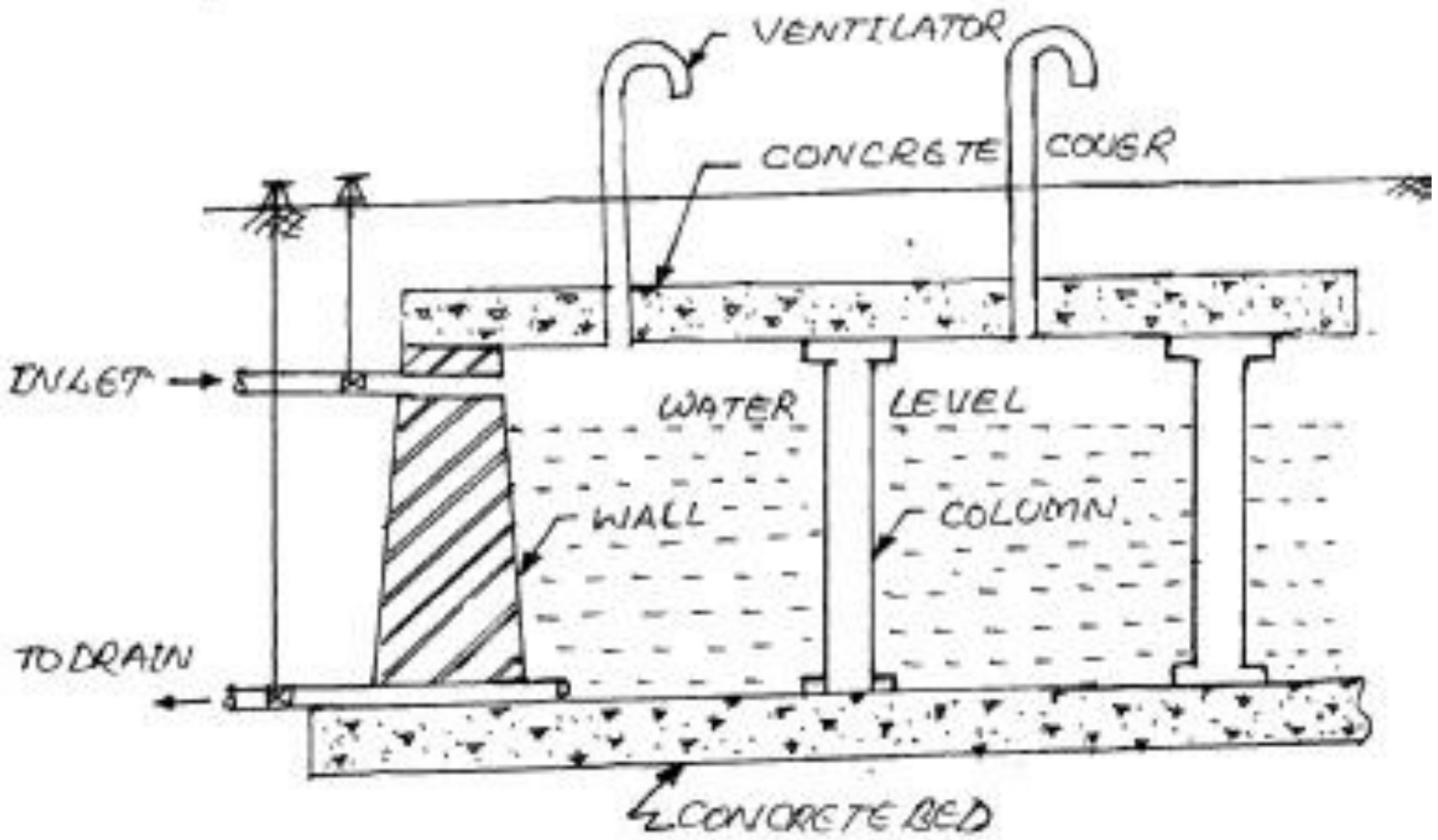
(SECTION OF GROUND RESERVOIR)

- The side walls are designed to take up the pressure of the water, when the reservoir is full and the earth pressure when it is empty.

- The position of ground water table is also considered while designing these reservoirs.

- The floors of these reservoirs may constructed with R.C.C slab or square stone blocks resting on columns.

- To obtain water tightness bitumen compounds are used at all construction joints.
- At the top of roof about 60cm thick earth layer is deposited and maintained green lawns to protect the reservoir from cold and heat.
- For aeration of water and inspection, ventilation pipes and stairs are provided.



Under Ground Reservoir

TYPES OF TANKS

R.C.C TANKS: R.C.C tanks are very popular because

1) They have long life
2) Very little maintenance
3) decent appearance

G.I. TANKS: G.I. tanks are generally in rectangular or square in shape. Now a days G.I. tanks are not preferring because

1) Life of the tank is short
2) Corrosion of metal
3) maintenance cost may be more

HDPE TANKS: Now a days HDPE tanks are very popular for storing less quantity of water and hence useful for residential purpose. The following are the advantages of HDPE tanks

1) Handling is easy because of light weight
2) Cheap in cost
3) Maintenance cost is low
4) Cleaning of tanks are easy

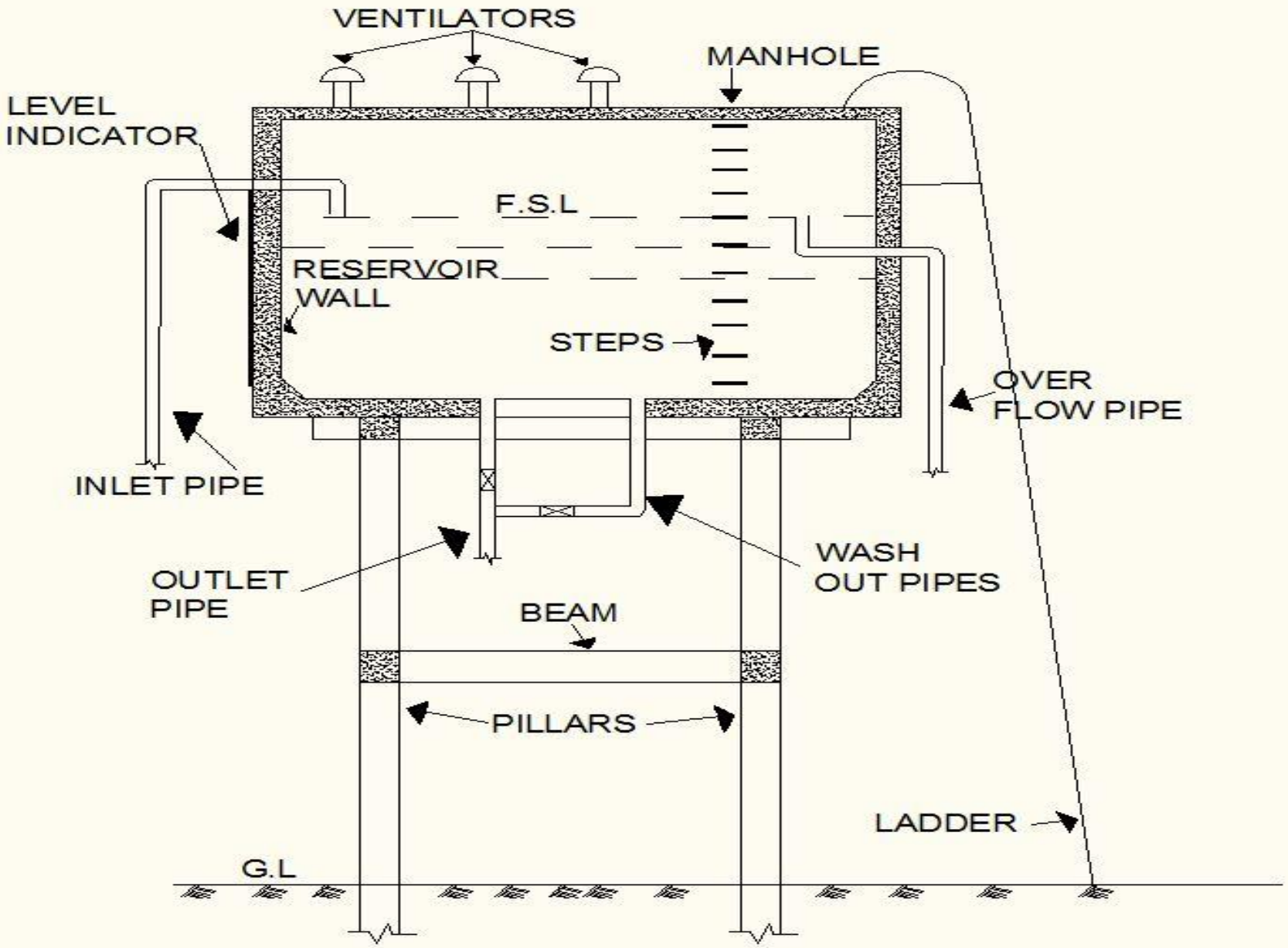
ESR...

- Elevated Storage Reservoirs (ESRs) also referred to as Overhead Tanks are required at distribution areas which are not governed and controlled by the gravity system of distribution.
- These are rectangular, circular or elliptical in shape.
- If the topography of the town not suitable for under gravity, the elevated tank or reservoir are used.

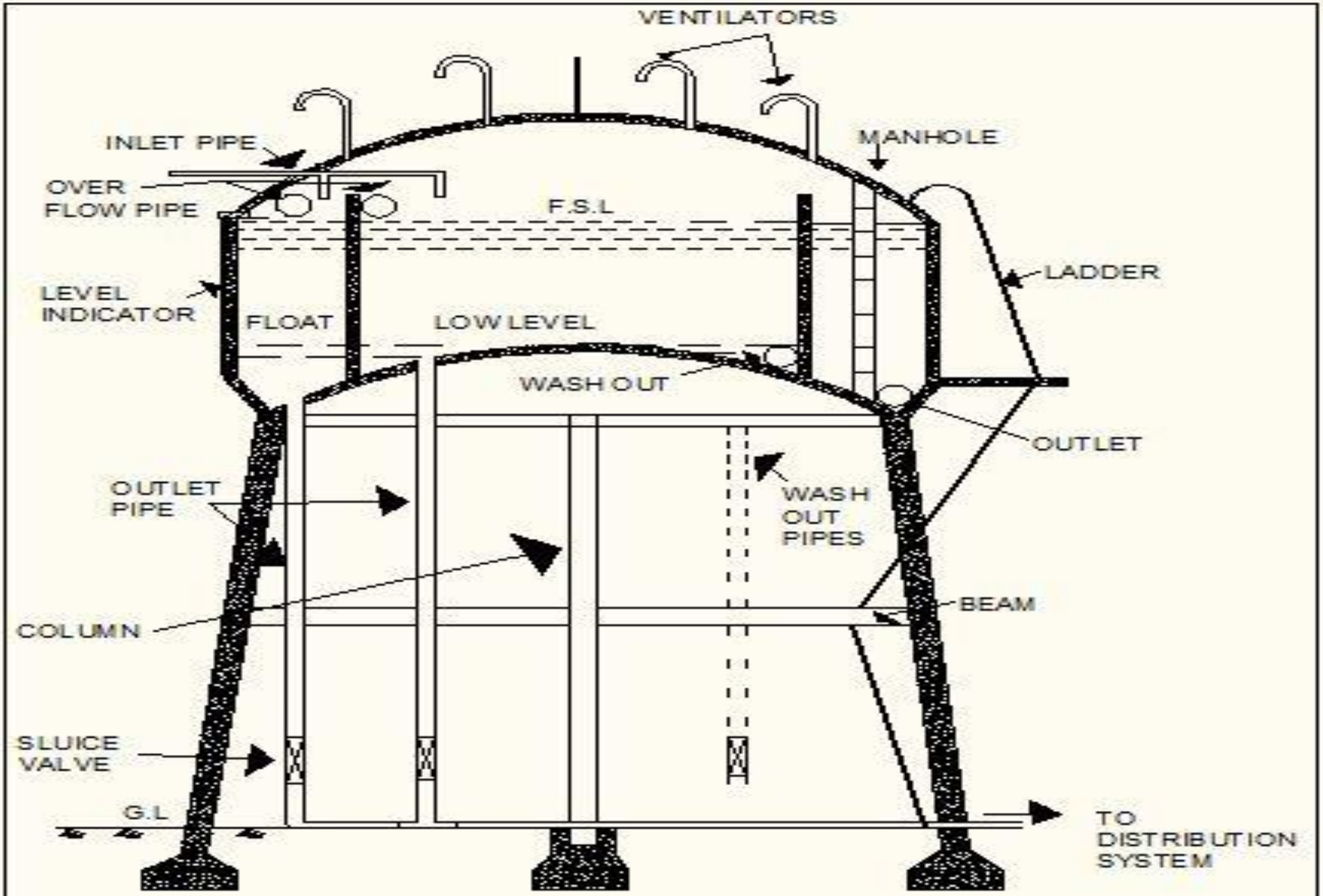
- They are constructed where combine gravity and pumping system of water distribution is adopted.
- These tanks may be steel or RCC.
- Now RCC is commonly preferred.

The accessories of ESR are-

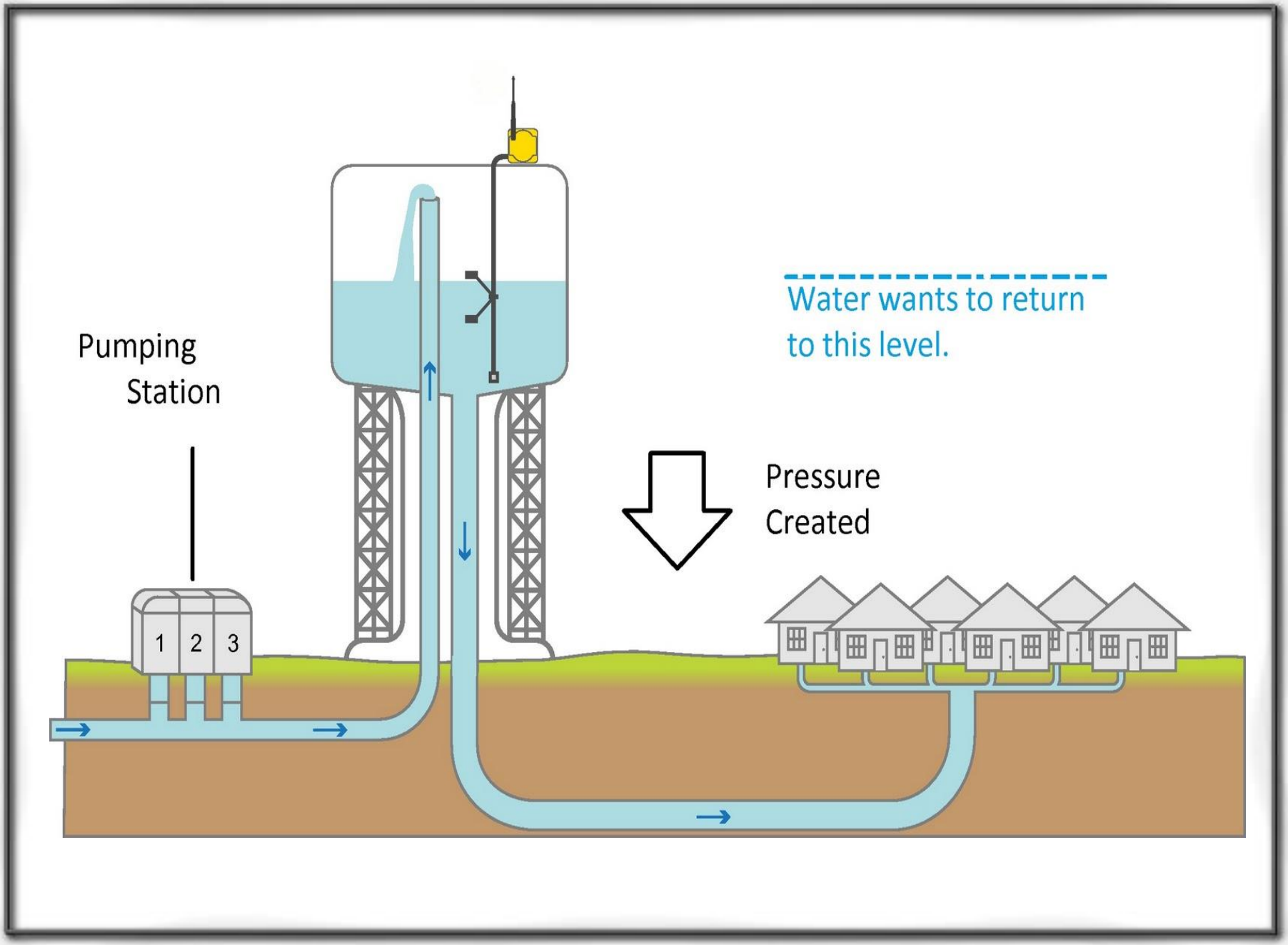
- Inlet and outlet pipe, overflow pipe discharging into a drain
- Float gauge, indicating depth of water.
- Automatic device to stop pumping when the tank is full.
- A manhole and ladder.
- Ventilator for circulation of fresh air.



(RECTANGULAR ELEVATED TANK)


VENTILATORS
INLET PIPE
MANHOLE
OVER
FLOW PIPE
F.S.L
LADDER
LEVEL
INDICATOR
FLOAT
LOW LEVEL
WASH OUT
OUTLET
OUTLET
PIPE
WASH
OUT
PIPES
BEAM
COLUMN
SLUICE
VALVE
G.L
TO
DISTRIBUTION
SYSTEM

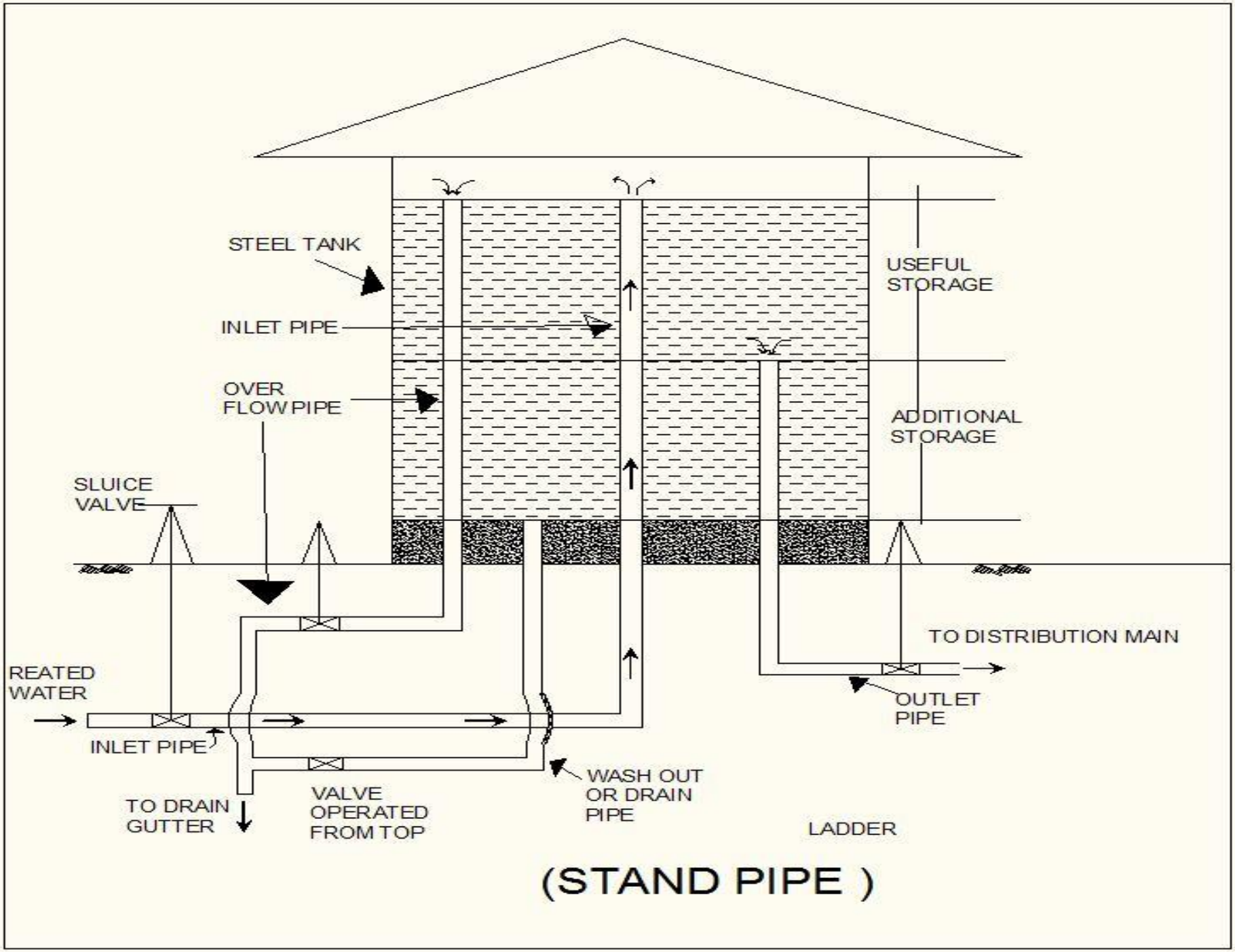
(INTZ TANK)


Pumping
Station
1
2
3
Water wants to return
to this level.
Pressure
Created

Stand Pipes

- Stand pipes are a kind of elevated tanks without any erected towers for resting the body.
- They are thus tall cylindrical shells resting directly on the ground.




STEEL TANK
INLET PIPE
OVER
FLOW PIPE
SLUICE
VALVE
USEFUL
STORAGE
ADDITIONAL
STORAGE
TO DISTRIBUTION MAIN
REATED
WATER
INLET PIPE
OUTLET
PIPE
TO DRAIN
GUTTER
VALVE
OPERATED
FROM TOP
WASH OUT
OR DRAIN
PIPE
LADDER

(STAND PIPE)

Storage Capacity

- The total storage capacity of a distribution reservoir is the summation of :
    1. Balancing storage (equalizing/operating storage)
    2. Break-down storage And
    3. Fire storage
- Balancing storage : The main and primary function of the distribution reservoir is to meet the fluctuating demand with a constant rate of supply from the treatment plant. The quantity of water required to be stored in the reservoir for equalizing or balancing this variable demand against the constant supply.

Storage Capacity of Distribution Reservoirs...

- The total storage capacity of a distribution reservoir is the summation of:
- ***Balancing Storage:*** The quantity of water required to be stored in the reservoir for equalising or balancing fluctuating demand against constant supply is known as the balancing storage (or equalising or operating storage).

- ***Breakdown Storage:*** The breakdown storage or often called emergency storage is the storage preserved in order to tide over the emergencies posed by the failure of pumps, electricity, or any other mechanism driving the pumps.
- A value of about 25% of the total storage capacity of reservoirs, or 1.5 to 2 times of the average hourly supply, may be considered as enough provision for accounting this storage.

- ***Fire Storage:*** The third component of the total reservoir storage is the fire storage.
- This provision takes care of the requirements of water for extinguishing fires.
- A provision of 1 to 4 per person per day is sufficient to meet the requirement.

- When reserve storage is elevated, amount of fire reserve may be determined by
- R= (F-P) T
- R= Reserve storage (liters)
- F= Fire demand, liters/min
- P= Reserve fire pumping capacity, liters/min
- T= Duration of the fire in min

The total reservoir storage can finally be worked out by adding all the three storages.

Network Layout

- Next step is to estimate pipe sizes on the basis of water demand and local code requirements.
- The pipes are then drawn on a digital map (using AutoCAD, for example) starting from the water source.
- All the components (pipes, valves, fire hydrants) of the water network should be shown on the lines.

Introduction…

- The purpose of distribution system is to deliver water to consumer with appropriate quality, quantity and pressure.

- Distribution system is used to describe collectively the facilities used to supply water from its source to the point of usage.

Requirements of Good Distribution System...

- Water quality should not get deteriorated in the distribution pipes.
- It should be capable of supplying water at all the intended places with sufficient pressure head.
- It should be capable of supplying the requisite amount of water during fire fighting.

- The layout should be such that no consumer would be without water supply, during the repair of any section of the system.

- All the distribution pipes should be preferably laid one metre away or above the sewer lines.

- It should be fairly water-tight as to keep losses due to leakage to the minimum.

Layouts of Distribution Network

- The distribution pipes are generally laid below the road pavements, and as such their layouts generally follow the layouts of roads.

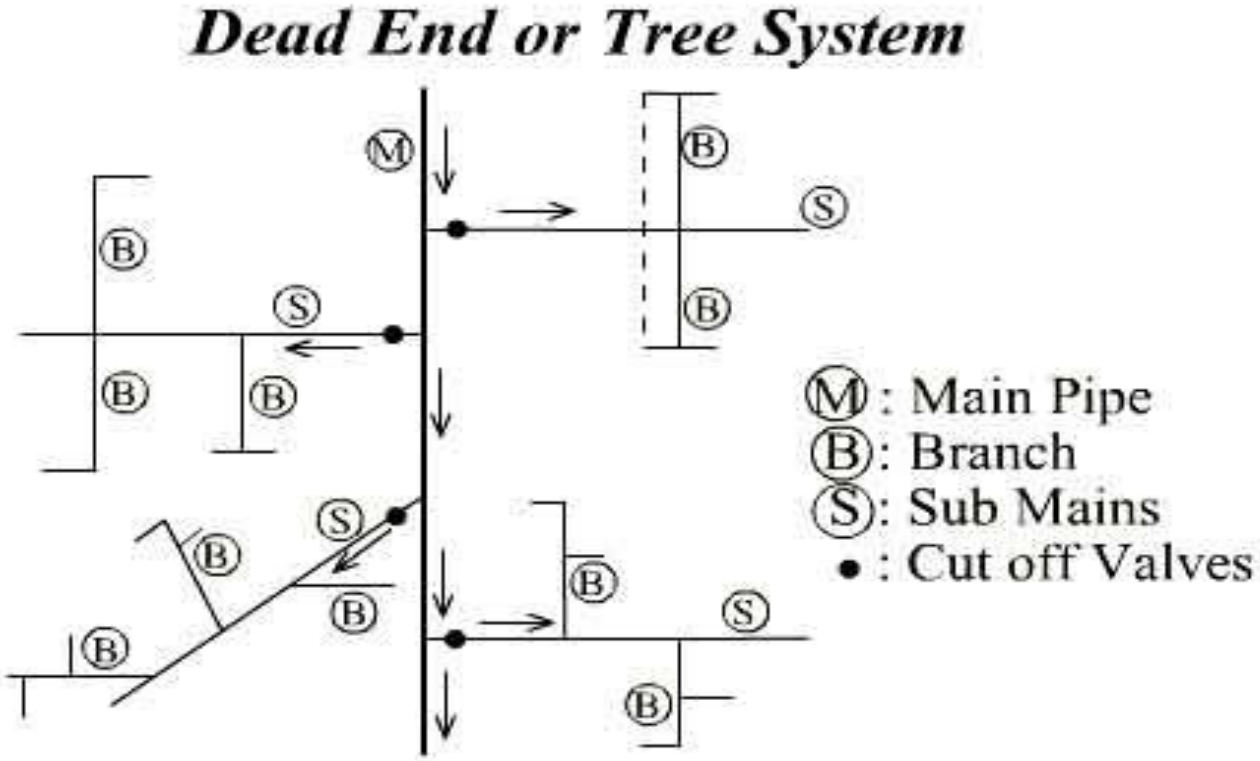
- There are, in general, four different types of pipe networks; any one of which either singly or in combinations, can be used for a particular place.

They are:

- ❖Dead End System
- ❖Radial System
- ❖Grid Iron System
- ❖Ring System

Dead End System...

- It is suitable for old towns and cities having no definite pattern of roads.



Advantages

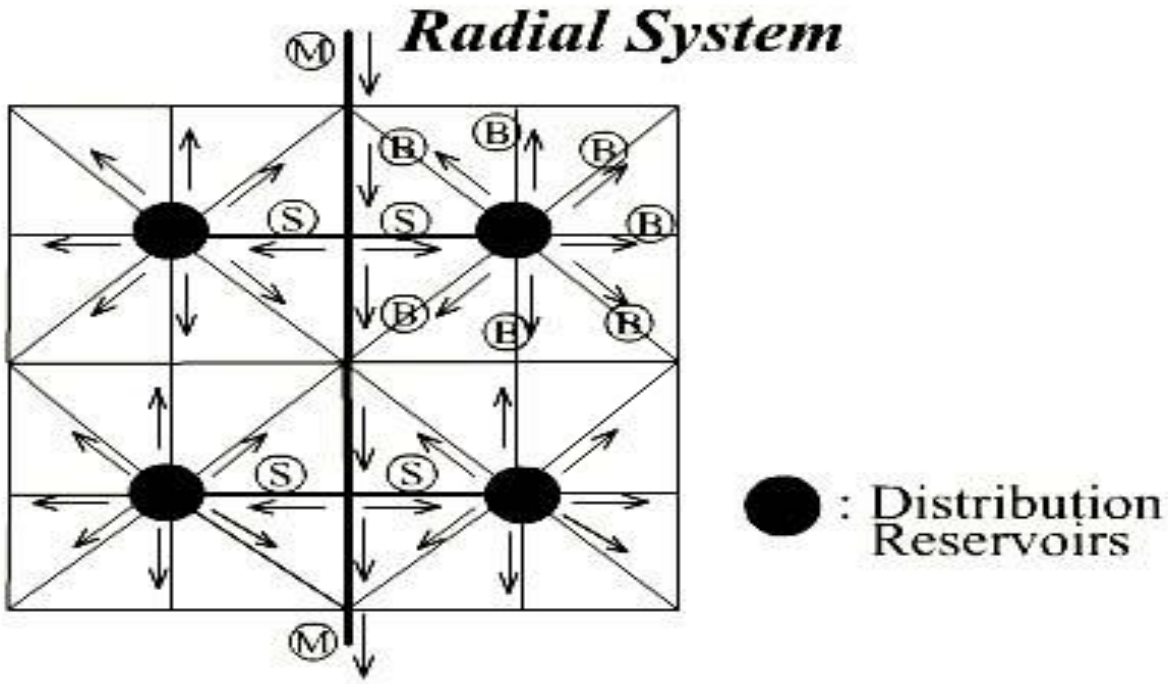
- Relatively cheap.
- Determination of discharges and pressure easier due to less number of valves.

Disadvantages

- Due to many dead ends, stagnation of water occurs in pipes.

Radial System...

- The area is divided into different zones.
- The water is pumped into the distribution reservoir kept in the middle of each zone.
- The supply pipes are laid radially ending towards the periphery.

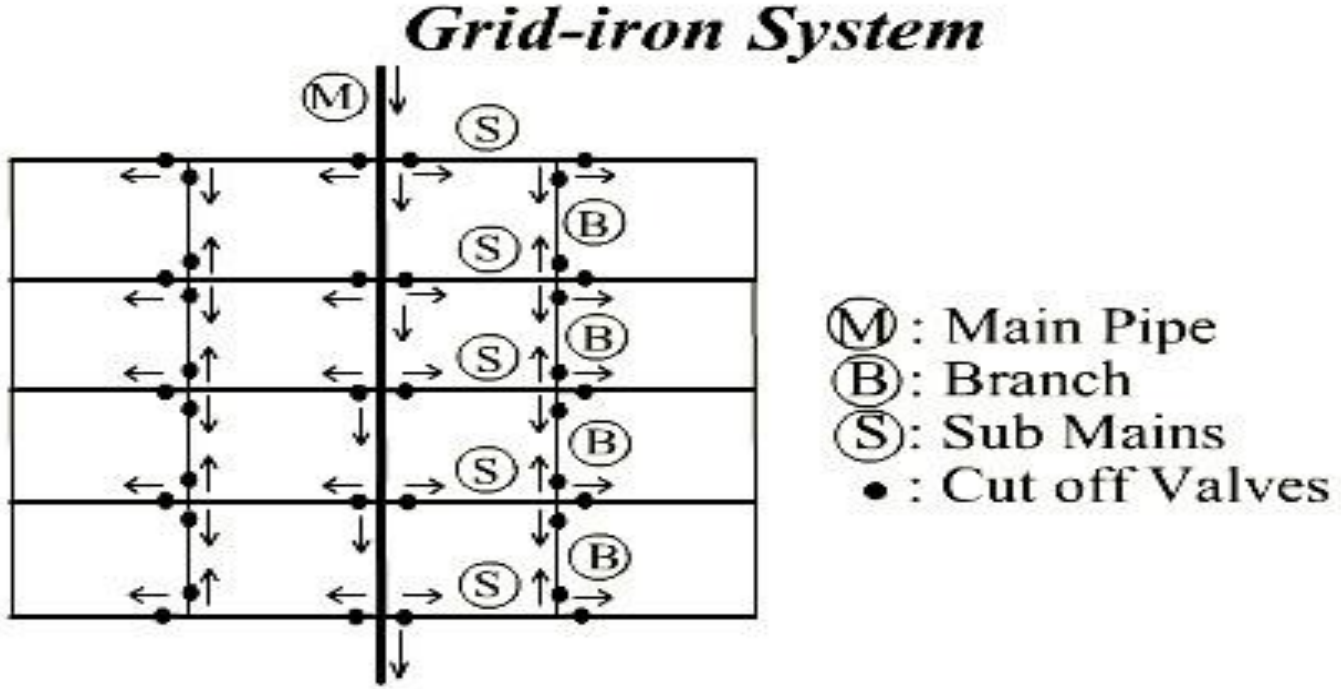


Advantages:

- It gives quick service.
- Calculation of pipe sizes is easy.

Grid Iron System...

- It is suitable for cities with rectangular layout, where the water mains and branches are laid in rectangles.



Advantages

- Water is kept in good circulation due to the absence of dead ends.
- In the cases of a breakdown in some section, water is available from some other direction.

Disadvantages

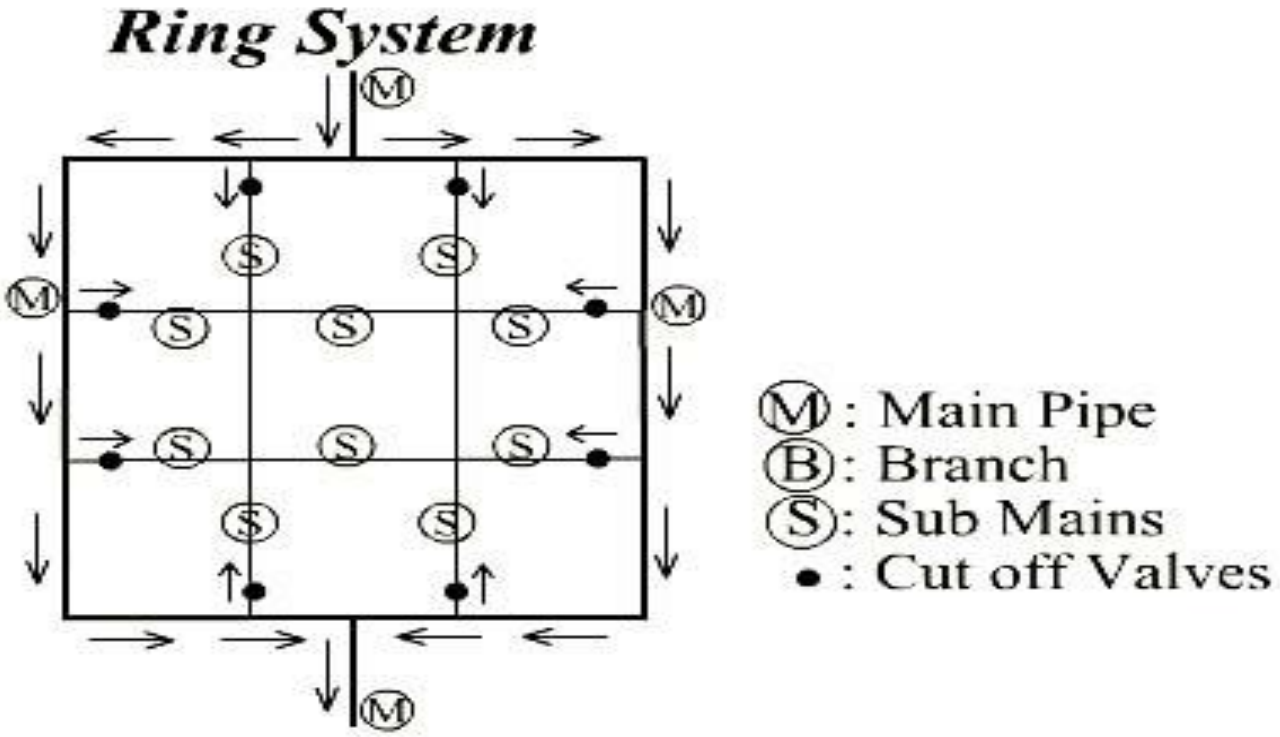
- Exact calculation of sizes of pipes is not possible due to provision of valves on all branches.

Ring System...

- The supply main is laid all along the peripheral roads and sub mains branch out from the mains.
- This system also follows the grid iron system with the flow pattern similar in character to that of dead end system.
- So, determination of the size of pipes is easy.

Advantages

- Water can be supplied to any point from at least two directions.



Methods of water distribution...

- For efficient distribution system adequate water pressure required at various points.

- Depending upon the level of source, topography of the area and other local conditions the water may be forced into distribution system by following ways -

1. Gravity system
2. Pumping system
3. Combined gravity and pumping system

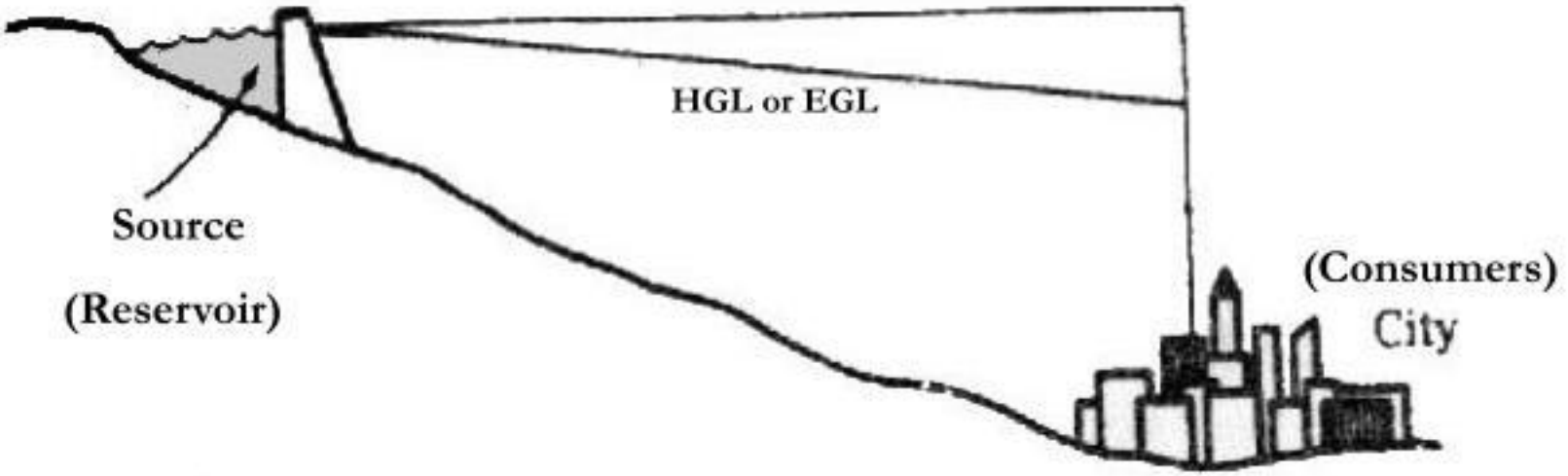
Gravity system…

- Suitable when source of supply is at sufficient height.
- Most reliable and economical distribution system.
- The water head available at the consumer is just minimum required.
- The remaining head is consumed in the frictional and other losses.

Gravity Supply

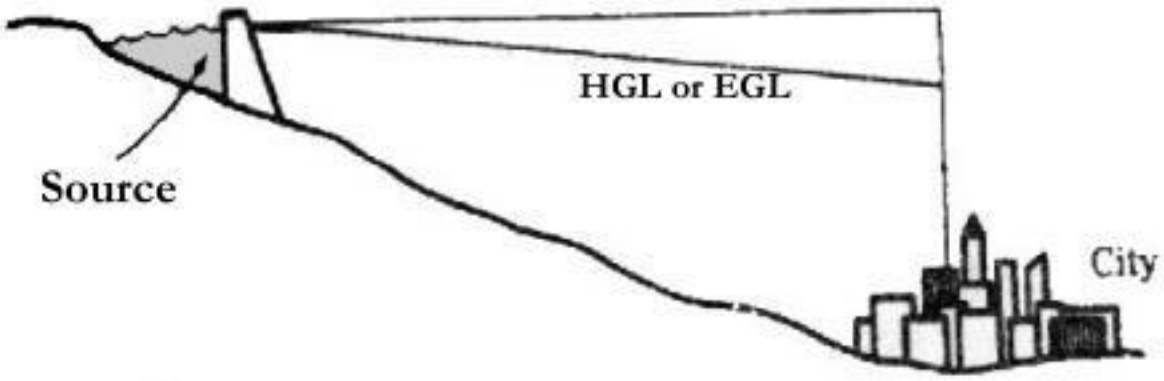
- The **source** of supply is at a sufficient elevation **above** the **distribution area** (consumers).

so that the desired pressure can be maintained

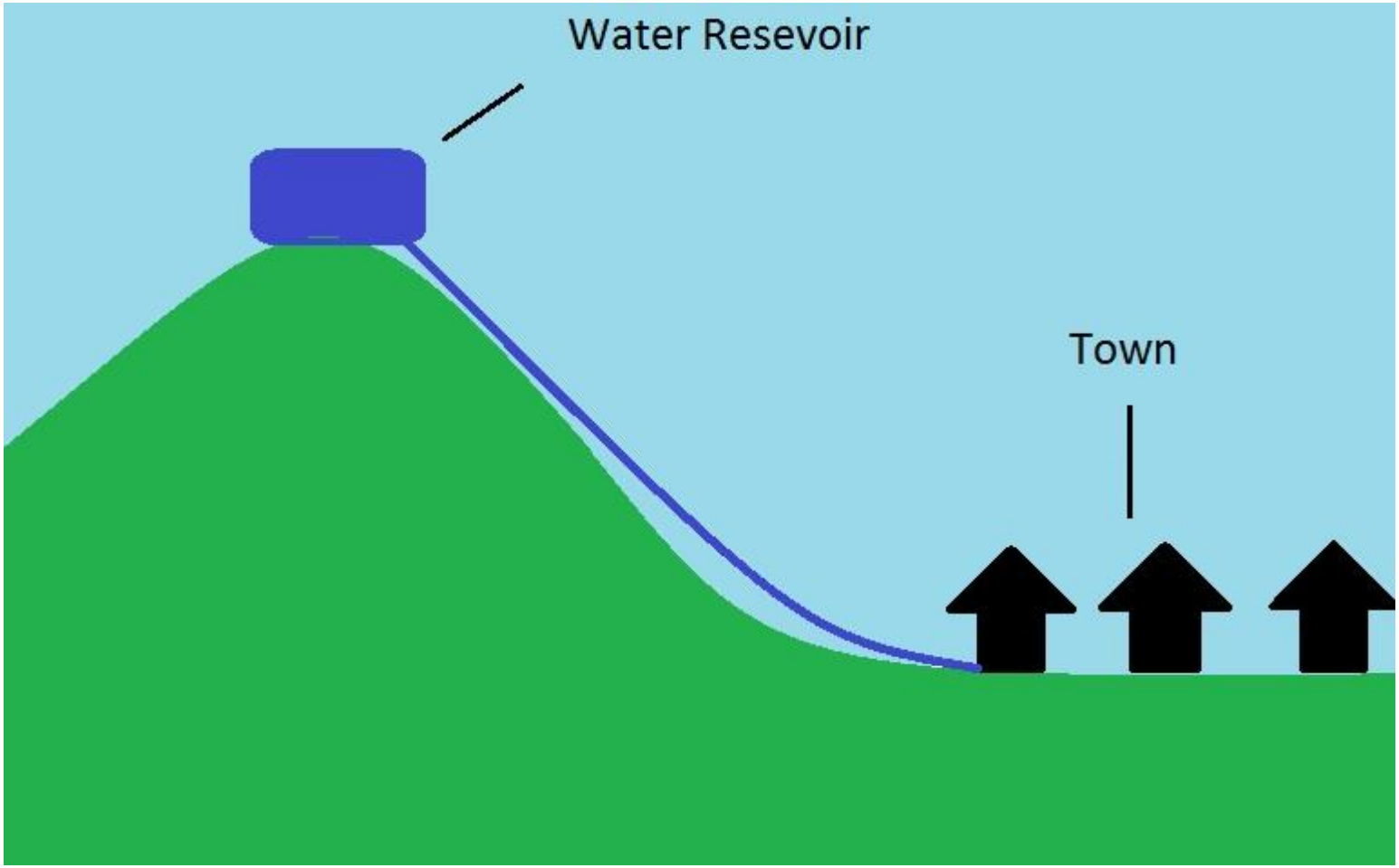


Gravity-Supply System

Advantages of Gravity supply

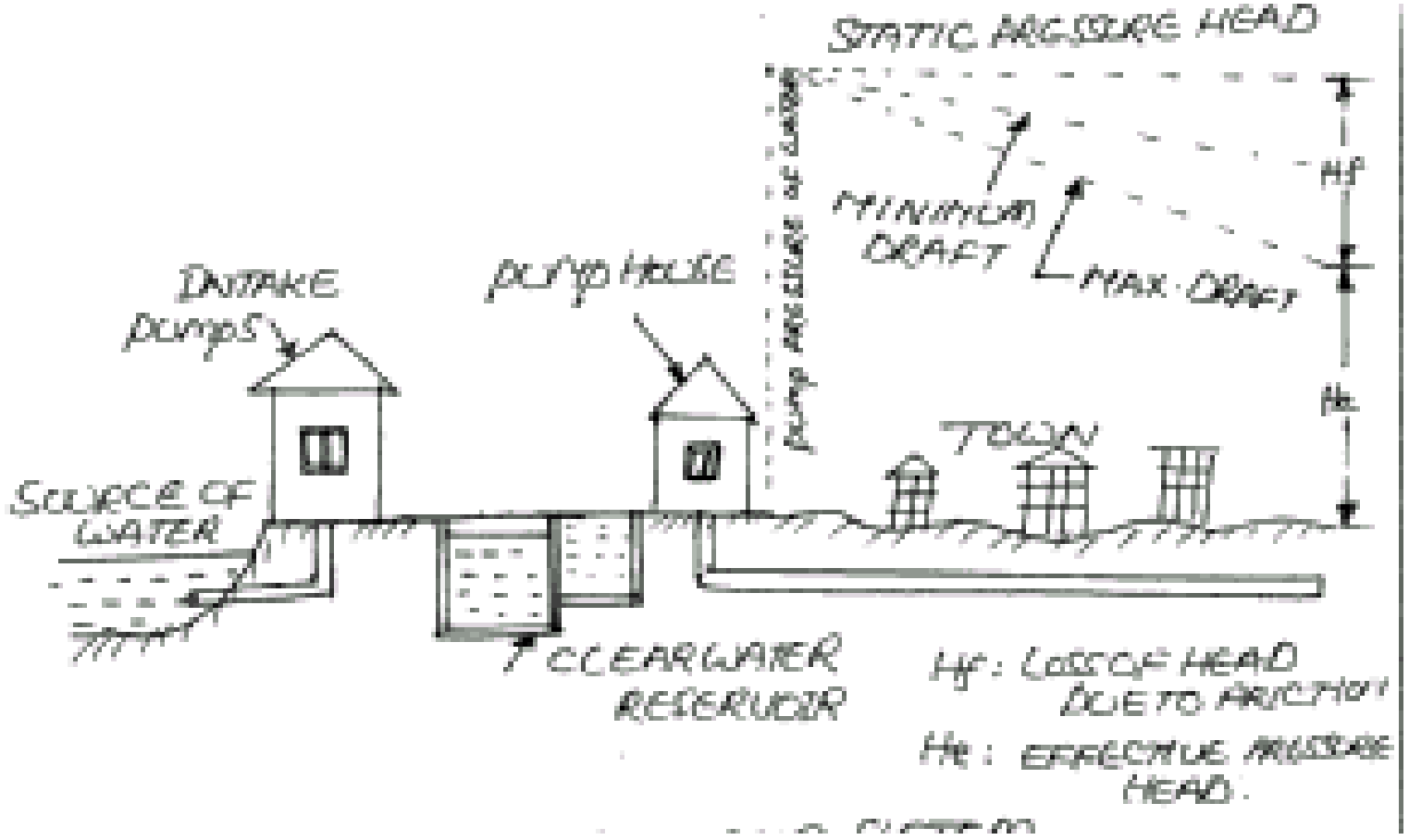


- No energy costs.
- Simple operation (fewer mechanical parts, independence of power supply, ….)
- Low maintenance costs.
- No sudden pressure changes


Water Resevoir
Town

Pumping system…

- Treated water is directly pumped in to the distribution main with out storing.
- Also called pumping without storage system.
- High lifts pumps are required.
- If power supply fails, complete stoppage of water supply.
- This method is not generally used.


STATIC PRESSURE HEAD
PUMP PRESSURE OR CURVE
MINIMUM DRAFT
MAX. DRAFT
Hf
He
INTAKE PUMPS
PUMP HOUSE
TOWN
SOURCE OF WATER
CLEAR WATER RESERVOIR
Hf : LOSS OF HEAD DUE TO FRICTION
He : EFFECTIVE PRESSURE HEAD.

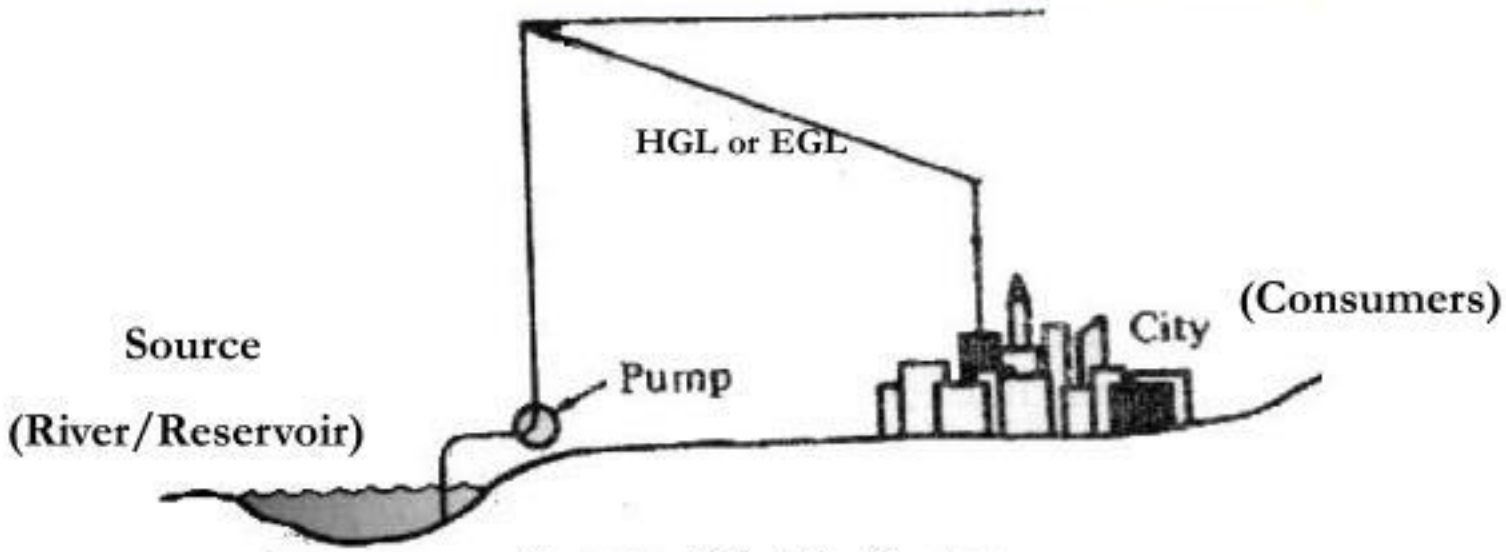
Water Distribution Systems

Pumped Supply

➔ Used whenever:

- The source of water is lower than the area to which we need to distribute water to (consumers)
- The source cannot maintain minimum pressure required.

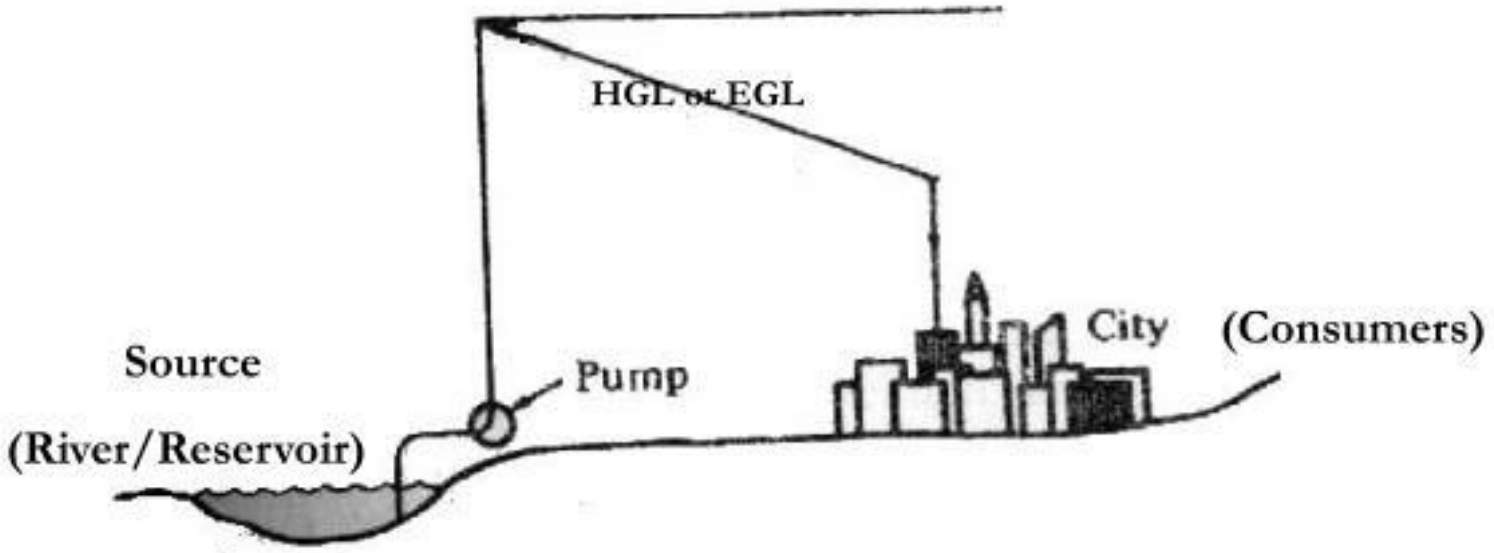
➔ pumps are used to develop the necessary head (pressure) to distribute water to the consumer and storage reservoirs.



Pumped-Supply System

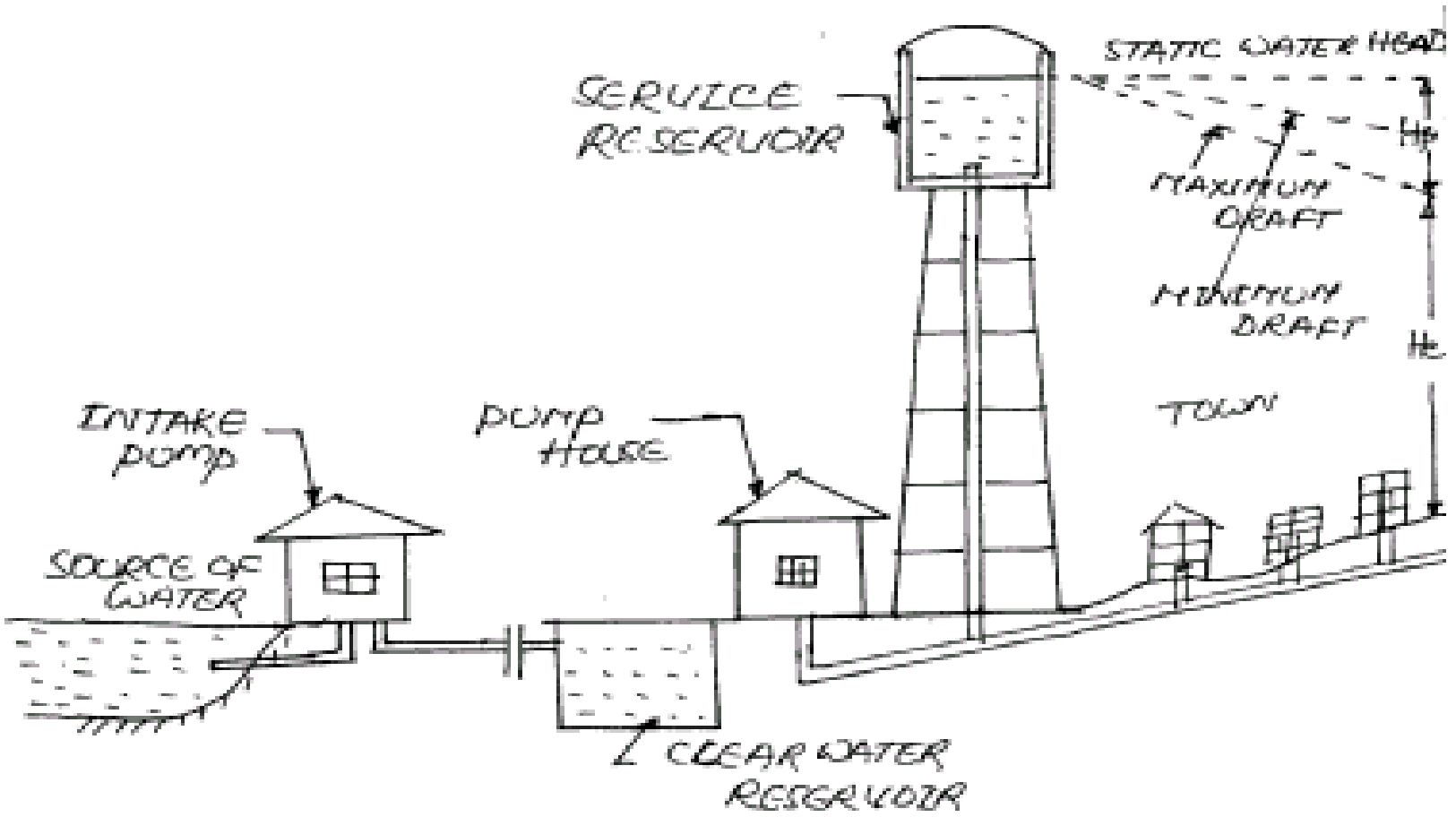
Disadvantages of pumped supply

- ☒ Complicated operation and maintenance.
- ☒ Dependent on reliable power supply.
- ☒ Precautions have to be taken in order to enable permanent supply:
  - Stock with spare parts
  - Alternative source of power supply ….



Combined gravity and pumping system

- Most common system.
- Treated water is pumped and stored in an elevated distribution reservoir.
- Then supplies to consumer by action of gravity.
- The excess water during low demand periods get stored in reservoir and get supplied during high demand period.
- Economical, efficient and reliable system.

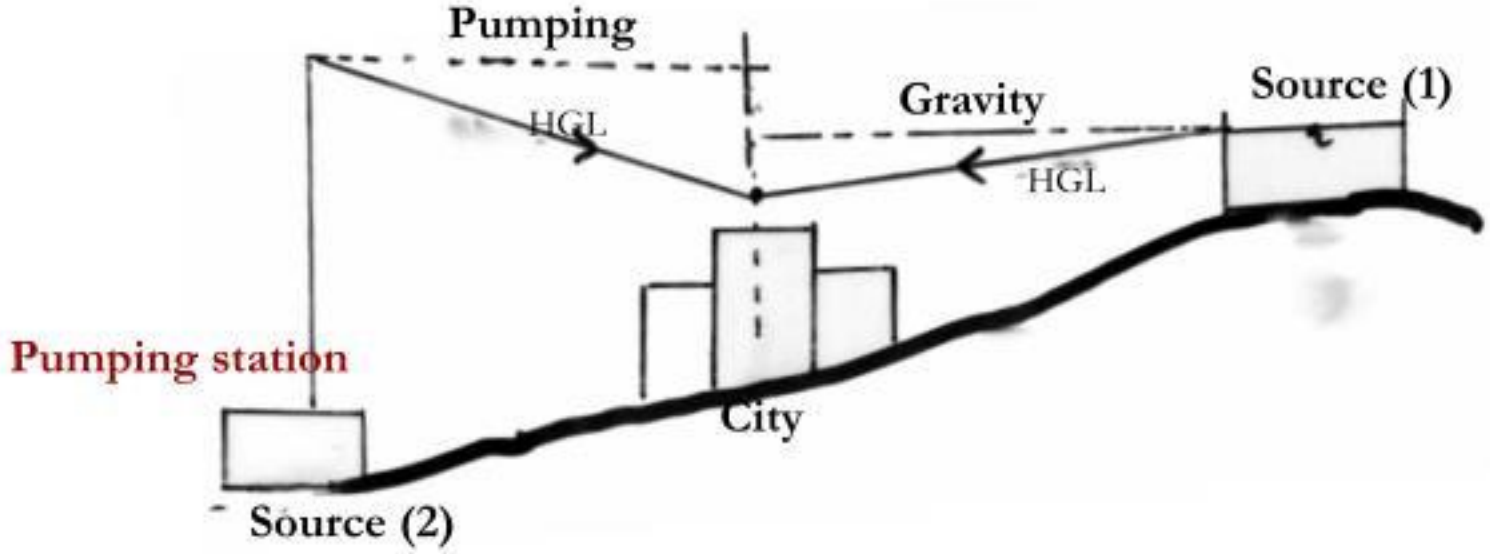

SERVICE RESERVOIR
STATIC WATER HEAD
Hs
MAXIMUM DRAFT
MINIMUM DRAFT
He
TOWN
INTAKE PUMP
PUMP HOUSE
SOURCE OF WATER
CLEAR WATER RESERVOIR

Combined Supply

(pumped-storage supply)

- Both **pumps** and **storage reservoirs** are used.
- This system is usually used in the following cases:

1) When two sources of water are used to supply water:

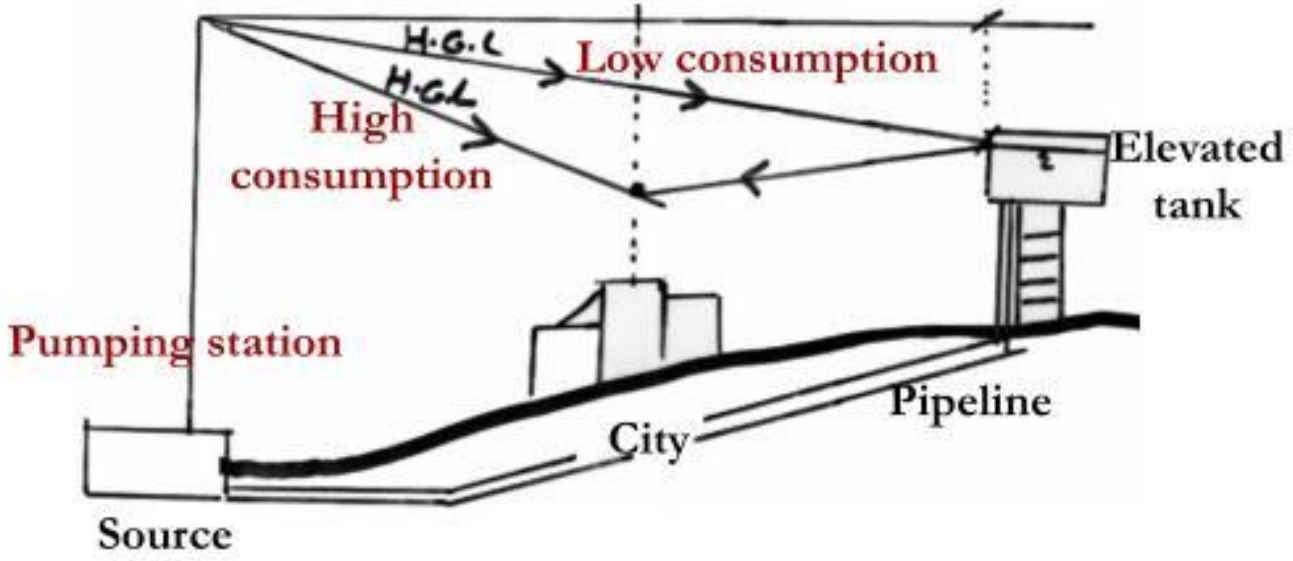


Water Distribution Systems

Combined Supply (Continue)

2) **In the pumped system sometimes a storage (elevated) tank is connected to the system.**

- When the water consumption is low, the residual water is pumped to the tank.
- When the consumption is high the water flows back to the consumer area by gravity.



Water Distribution Systems

Combined Supply (Continue)

3) When the source is lower than the consumer area

- A tank is constructed above the highest point in the area,
- Then the water is pumped from the source to the storage tank (reservoir).
- And the hence the water is distributed from the reservoir by gravity.

